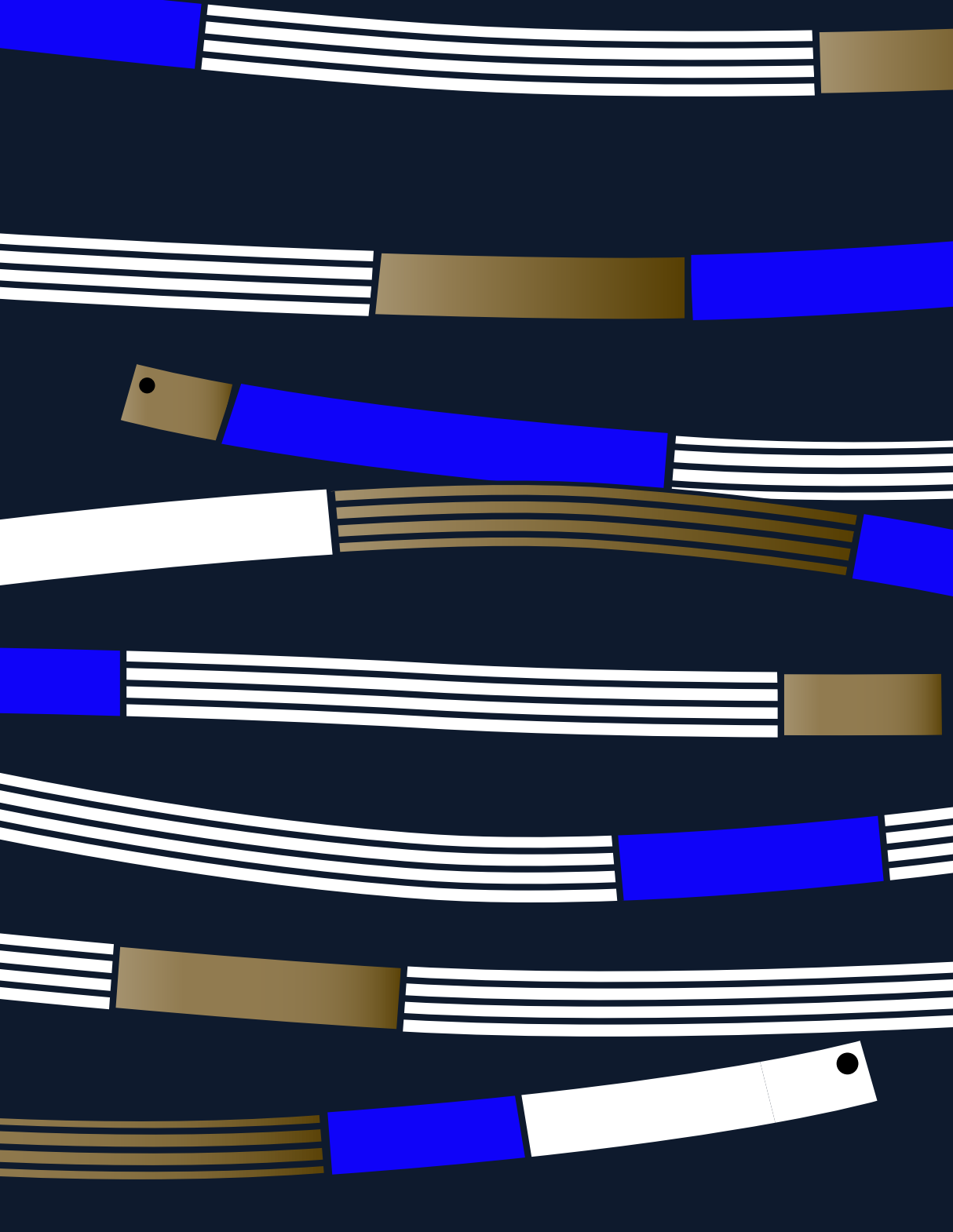




Objetivos:

1. **Presentar metodología de las clases de ayudantía.**
 2. **Resolver problemas que involucren sucesiones monótonas, acotadas y convergentes.**
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Parte 1: Información sobre ayudantía presencial de Cálculo Diferencial.

1. El ayudante se presenta e informa correo institucional como canal de comunicación con los estudiantes.
2. Se presenta el objetivo de estas clases “Reforzar los temas abordados en clases de cátedra a través de la resolución colaborativa de problemas para el fortalecimiento del trabajo autónomo de los estudiantes”.
3. Las ayudantías se llevarán a cabo semanalmente. En caso de ser necesario suspender una ayudantía, esta se informará vía canvas con antelación y se recuperará de forma presencial.
4. La asistencia es obligatoria y se requiere de un 70% como mínimo para aprobar el curso (9 clases como mínimo). El registro de asistencia se debe realizar a través de canvas.
5. En las ayudantías se realizarán diversas actividades enfocadas a reforzar los temas de cátedra abordados en la semana anterior.
6. No se aplicarán evaluaciones en las sesiones de ayudantía.

Parte 2: Resolución de problemas en conjunto con los estudiantes

El ayudante deberá resolver los siguientes problemas en pizarra, incentivando la participación de los estudiantes y reforzando los contenidos necesarios para abordar estos ejercicios.

Problema 1. Demuestre que la sucesión $a_n = \frac{n+8}{2+n}$ es monótona y acotada.

Solución:

Monotonía:

Recordar a los estudiantes que si:

$a_n - a_{n+1} \geq 0$ La sucesión es monótona decreciente (no estricta)

$a_n - a_{n+1} > 0$ La sucesión es monótona decreciente (estricta)

$a_n - a_{n+1} \leq 0$ La sucesión es monótona creciente (no estricta)

$a_n - a_{n+1} < 0$ La sucesión es monótona creciente (estricta)

$$a_m = \frac{m+8}{2+m}, \quad a_{m+1} = \frac{m+9}{3+m} \quad (a_{m+1} \leq a_m)$$

$$\frac{m+9}{3+m} \leq \frac{m+8}{2+m}$$

$$2m + 18 + m^2 + 9m \leq 3m + 24 + m^2 + 8m$$

$$0 \leq 6$$

(Verdad)

Entonces es monótona, decrece

$$\lim_{m \rightarrow +\infty} \frac{m+8}{2+m} \Rightarrow \frac{\frac{m}{m} + \frac{8}{m}}{\frac{2}{m} + \frac{m}{m}} \Rightarrow 1 \quad \text{converge}$$

$$\lim_{m \rightarrow -\infty} \frac{\frac{-m}{m} + \frac{8}{m}}{\frac{2}{m} + \frac{-m}{m}} \Rightarrow 1 \quad \text{converge}$$

$\therefore$ concluimos que es acotada convergente

Veamos en este caso:

$$a_n - a_{n+1} = \frac{n+8}{2+n} - \frac{n+9}{3+n} = \frac{(n+8)(3+n) - (n+9)(2+n)}{(2+n)(3+n)} = \frac{6}{(2+n)(3+n)}$$

Como es claro que $\frac{6}{(2+n)(3+n)} > 0, \forall n \in \mathbb{N}$, entonces a_n es monótona decreciente estricta.

Observación: si se pide demostrar la monotonía no se puede comenzar asumiendo que la sucesión es creciente o decreciente.

Acotamiento:

Recordar a los estudiantes que:

a_n es acotada si $|a_n| \leq M, M \in \mathbb{R}$

Veamos en este caso:

$$a_n = \frac{(n+8) \cdot \frac{1}{n}}{(2+n) \cdot \frac{1}{n}} = \frac{1 + \frac{8}{n}}{\frac{2}{n} + 1}$$

Analizando el numerador, se tiene que:

$$\left| 1 + \frac{8}{n} \right| \leq |1| + \left| \frac{8}{n} \right| \leq 1 + \frac{8}{n} \leq 9, \forall n \in \mathbb{N}$$

Analizando el denominador, se tiene que:

$$\left| \frac{2}{n} + 1 \right| \leq \left| \frac{2}{n} \right| + |1| \leq \frac{2}{n} + 1 \leq 3, \forall n \in \mathbb{N}$$

Luego, $\left| \frac{n+8}{2+n} \right| \leq \frac{9}{3} \leq 3, \forall n \in \mathbb{N}$

Problema 2. Determine si la sucesión $a_n = \left(\frac{1+n}{2n+1} \right)^{-2}$ es convergente.

Recordar a los estudiantes que la sucesión es convergente si:

$$\lim_{n \rightarrow \infty} a_n = L \text{ con } L \in \mathbb{R}$$

Veamos en este caso:

Problema 2. Determine si la sucesión $a_n = \left(\frac{1+n}{2n+1}\right)^{-2}$ es convergente.

$$a_n = \left(\frac{2n+1}{1+n}\right)^2$$

$$\left(\lim_{n \rightarrow \infty} \frac{2n+1}{1+n}\right)^2$$

$$\left(\lim_{n \rightarrow \infty} \frac{\frac{2n}{n} + \frac{1}{n}}{\frac{1}{n} + \frac{n}{n}}\right)^2$$

$$\left(\lim_{n \rightarrow \infty} \frac{2}{1}\right)^2 \Rightarrow \lim_{n \rightarrow \infty} 4 \quad \text{converge}$$

$$\lim_{n \rightarrow \infty} \left(\frac{1+n}{2n+1} \right)^{-2} = \left[\lim_{n \rightarrow \infty} \left(\frac{1+n}{2n+1} \right) \right]^{-2} = \left[\lim_{n \rightarrow \infty} \left(\frac{n \left(\frac{1}{n} + 1 \right)}{n \left(2 + \frac{1}{n} \right)} \right) \right]^{-2} = \left[\lim_{n \rightarrow \infty} \left(\frac{\frac{1}{n} + 1}{2 + \frac{1}{n}} \right) \right]^{-2} = \left[\frac{1}{2} \right]^{-2} = 4$$

Problema 3. Determine si la sucesión $\left\{ \frac{n+2}{8n-10} \right\}$ es monótona y convergente.

Monotonía:

$$\frac{n+2}{8n-10} - \frac{n+3}{8n-2} = \frac{(n+2)(4n-1) - (n+3)(4n-5)}{2(4n-5)(4n-1)} = \frac{13}{2(4n-5)(4n-1)}$$

Como $\frac{13}{2(4n-5)(4n-1)} > 0$ si $n \geq 2$ y $\frac{13}{2(4n-5)(4n-1)} < 0$ si $n < 2$, entonces la sucesión no es monótona.

Convergencia:

$$\lim_{n \rightarrow \infty} \frac{(n+2) \cdot \frac{1}{n}}{(8n-10) \cdot \frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{8 - \frac{10}{n}} = \frac{1}{8}$$

La sucesión es convergente.

Problema 4. Se sabe que la sucesión $a_n = \sqrt[3]{\frac{n^2-1}{5-an^2}}$ con $a \in \mathbb{R}$ converge en $-\frac{1}{2}$, determine el valor de a .

$$\lim_{n \rightarrow \infty} \sqrt[3]{\frac{n^2-1}{5-an^2}} = -\frac{1}{2}$$

$$\sqrt[3]{\lim_{n \rightarrow \infty} \frac{n^2 \left(1 - \frac{1}{n^2} \right)}{n^2 \left(\frac{5}{n^2} - a \right)}} = -\frac{1}{2}$$

$$\sqrt[3]{\lim_{n \rightarrow \infty} \frac{1 - \frac{1}{n^2}}{\frac{5}{n^2} - a}} = -\frac{1}{2}$$

$$\sqrt[3]{-\frac{1}{a}} = -\frac{1}{2}$$

$$a = 8$$

Problema 3. Determine si la sucesión $\left\{ \frac{n+2}{8n-10} \right\}$ es monótona y convergente.

$$a_{n+1} = \frac{n+3}{8n+14}$$

$$\frac{n+2}{8n-10} \leq \frac{n+3}{8n+14}$$

$$\frac{n+2}{8n-10} - \frac{n+3}{8n+14} \leq 0$$

$$\cancel{8n^2} + 16n + \cancel{14n} - 4 - \cancel{8n^2} - 10n + 24n - 30 \leq 0$$

$$\frac{26}{26}$$

$$64n^2 - 16n - 8n + 20 \quad \therefore \text{no es monótona}$$

EJERCICIOS CONTROL

1.-

Determine si la sucesión $a_n = \left(\frac{1+n}{2n+1}\right)^{-2}$

- a) Es monótona
- b) Convergente
- c) Acotada

2.-

Considere la sucesión recursiva convergente con $a_n > 0$ definida como:

$$a_1 = 3, a_{n+1} = \frac{1}{2} \left(a_n + \frac{25}{a_n} \right)$$

a. Encuentre a_3 .

b. Calcule $\lim_{n \rightarrow \infty} a_n$

Solución:

$$a. a_2 = \frac{1}{2} \left(a_1 + \frac{25}{a_1} \right) = \frac{1}{2} \left(3 + \frac{25}{3} \right) = \frac{17}{3} \rightarrow a_3 = \frac{1}{2} \left(a_2 + \frac{25}{a_2} \right) = \frac{1}{2} \left(\frac{17}{3} + \frac{25}{\frac{17}{3}} \right) = \frac{257}{51}$$

$$b. \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_{n+1} = L$$

$$L = \lim_{n \rightarrow \infty} \frac{1}{2} \left(a_n + \frac{25}{a_n} \right)$$

$$L = \frac{1}{2} \left(L + \frac{25}{L} \right)$$

$$L^2 = 25 \rightarrow L = 5 \vee L = -5$$

Como $a_n > 0$, entonces $\lim_{n \rightarrow \infty} a_n = 5$

1.-

Determine si la sucesión $a_n = \left(\frac{1+n}{2n+1}\right)^{-2}$

- a) Es monótona
- b) Convergente
- c) Acotada

$$a_n = \left(\frac{2n+1}{1+n}\right)^2, \quad a_{n+1} = \left(\frac{2n+3}{n+2}\right)^2$$

$$\frac{4n^2 + 4n + 1}{n^2 + 2n + 1} < \frac{4n^2 + 12n + 9}{n^2 + 4n + 4}$$

$$(4n^2 + 4n + 1)(n^2 + 4n + 4) < (4n^2 + 12n + 9)(n^2 + 2n + 1)$$

$$4n^4 + 16n^3 + 16n^2 + 4n^3 + 16n^2 + 16n + n^2 + 4n + 4 < 4n^4 + 8n^3 + 4n^3 + 12n^2 + 24n^2 + 12n + 9n^2 + 18n + 9$$

$$-5 < 4n^2 + 10n \quad (\text{verdadero})$$

$\therefore$ es monótona, Positivamente

$$b) \left[\lim_{n \rightarrow \infty} \frac{1+n}{2n+1} \right]^{-2}$$

$$\left[\lim_{n \rightarrow \infty} \frac{2n+1}{1+n} \right]^{-2} \Rightarrow 4 \quad \text{Converge}$$

$$c) a_n = \left(\frac{3}{2}\right)^2 \Rightarrow \frac{9}{4}$$

acotada sup: 4
cotu Inf: 9/4

Considere la sucesión recursiva convergente con $a_n > 0$ definida como:

$$a_1 = 3, a_{n+1} = \frac{1}{2} \left(a_n + \frac{25}{a_n} \right)$$

a. Encuentre a_3 .

b. Calcule $\lim_{n \rightarrow \infty} a_n$

$$a_1 = 3; \quad a_{n+1} = \frac{1}{2} \left(a_n + \frac{25}{a_n} \right)$$

$$a_2 = \frac{1}{2} \left(3 + \frac{25}{3} \right) \Rightarrow \frac{17}{3} \cdot \frac{1}{2} \Rightarrow \frac{17}{3} \approx 5.6$$

$$a_3 = \frac{1}{2} \left(\frac{17}{3} + \frac{25}{\frac{17}{3}} \right)$$

$$a_3 = \frac{1}{2} \left(\frac{17}{3} + \frac{25 \cdot 3}{17} \right)$$

$$a_3 = \frac{1}{2} \left(\frac{314}{17} \right) \Rightarrow a_3 = \frac{157}{17}$$

$$b) \lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(a_n + \frac{25}{a_n} \right)$$

$$\lim_{n \rightarrow \infty} 2L = \lim_{n \rightarrow \infty} L + \frac{25}{L}$$

$$2L = L + \frac{25}{L}$$

$$L = \frac{25}{L}$$

$$L^2 = 25$$

$$L = 5 \quad \vee \quad L = -5$$

~~$L = -5$~~
 $a_n > 0$

3.-

Estudie la convergencia de la sucesión $\{\sqrt{n+3} - \sqrt{n}\}$.

Estudie la convergencia de la sucesión $\{\sqrt{n+3} - \sqrt{n}\}$.

$$\lim_{n \rightarrow \infty} \sqrt{n+3} - \sqrt{n} \Rightarrow \lim_{n \rightarrow \infty} \sqrt{n+3} - \sqrt{n} \cdot \frac{\sqrt{n+3} + \sqrt{n}}{\sqrt{n+3} + \sqrt{n}}$$

$$\lim_{n \rightarrow \infty} \frac{n+3-n}{\sqrt{n+3} + \sqrt{n}} \Rightarrow \frac{3}{\sqrt{n+3} + \sqrt{n}} \Rightarrow \frac{\sqrt{n} \left(\frac{3}{\sqrt{n}} \right)}{\sqrt{n} \left(\sqrt{1 + \frac{3}{n}} + 1 \right)} \Rightarrow 0$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{x^2+9}-3}{x^2} \Rightarrow \lim_{x \rightarrow 0} \frac{\sqrt{x^2+9}-3}{x^2} \cdot \frac{\sqrt{x^2+9}+3}{\sqrt{x^2+9}+3} \Rightarrow \lim_{x \rightarrow 0} \frac{\cancel{x^2}}{\cancel{x^2}(\sqrt{x^2+9}+3)} \Rightarrow \frac{1}{6}$$

$$\lim_{x \rightarrow 0} \left(x^3 + \frac{\cos(5x)}{10000} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{10000} \right) \Rightarrow 0,0001$$

EJERCICIOS TALLER

1.- Sea $a_n = \frac{n+1}{n}$

- a) Determine si a_n es monótona.
- b) Muestre que a_n es acotada y convergente.

2.- Sea la función:

$$g(x) = \begin{cases} x + 5; & x \leq -3 \\ \sqrt{9 - x^2}; & -3 < x < 3 \\ 3 - x; & x \geq 3 \end{cases}$$

Estudiar si existen los límites de la función en $x = 3, x = -3$

3.- Calcular

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2} \right) =$$

4.- Calcular

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 6x} - (x - 3)}{x + 3 - \sqrt{x^2 + 6x}}$$

1.- Sea $a_n = \frac{n+1}{n}$

a) Determine si a_n es monótona.

b) Muestre que a_n es acotada y convergente.

a) $a_{n+1} = \frac{n+2}{n+1}$

Secuencia: $2, \frac{3}{2}, \frac{4}{3} \rightarrow$

$$\frac{n+1}{n} > \frac{n+2}{n+1}$$

$$(n+1)^2 > n^2 + 2n$$

$$\cancel{n^2 + 2n} + 1 > \cancel{n^2 + 2n}$$

$$1 > 0 \quad \checkmark \checkmark$$

es monótona decreciente

2) $\lim_{n \rightarrow \infty} \frac{n+1}{n} \Rightarrow \lim_{n \rightarrow \infty} \frac{\cancel{n} + \frac{1}{\cancel{n}}}{\cancel{n}}$ $\Rightarrow 1$

∴ Dado que a_n es monótona y converge, es acotada

cota sup: 2, cota inf: 1

2.- Sea la función:

$$g(x) = \begin{cases} x+5; & x \leq -3 \\ \sqrt{9-x^2}; & -3 < x < 3 \\ 3-x; & x \geq 3 \end{cases}$$

$$\left. \begin{array}{l} \lim_{x \rightarrow -3^-} x+5 \Rightarrow 2 \\ \lim_{x \rightarrow -3^+} \sqrt{9-x^2} \Rightarrow 0 \end{array} \right\} \text{Lim no existe}$$

$$\left. \begin{array}{l} \lim_{x \rightarrow 3^-} \sqrt{9-x^2} \Rightarrow 0 \\ \lim_{x \rightarrow 3^+} 3-x \Rightarrow 0 \end{array} \right\} \text{Lim existe}$$

∴ el lim de la función en el intervalo $x \leq -3$ y $-3 < x$ no existe
 Pero el lim de la función en el intervalo $x < 3$ y $x \geq 3$ si existe
 límites laterales distintos por lo tanto el lim en la función no existe

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = 1^\infty \neq I$$

$$\lim_{x \rightarrow 0} e^{\ln(1+x)^{\frac{1}{x}}} \Rightarrow \lim_{x \rightarrow 0} \frac{1}{x} \ln(1+x) \Rightarrow \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \frac{0}{0} \Rightarrow \lim_{t \rightarrow 0} \frac{t = e^x - 1}{t+1 = e^x} \Rightarrow \lim_{t \rightarrow 0} \frac{t}{\ln(t+1)} \cdot \frac{\frac{1}{t}}{\frac{1}{t}} \Rightarrow \lim_{t \rightarrow 0} \frac{1}{\frac{1}{t} \ln(t+1)} \Rightarrow \lim_{t \rightarrow 0} \frac{1}{\ln(t+1)^{1/4}}$$

$x = \ln(t+1)$
 $t = 0$

$$\frac{\lim_{t \rightarrow 0} 1}{\lim_{t \rightarrow 0} \ln(t+1)^{1/4}} \Rightarrow \frac{1}{\lim_{t \rightarrow 0} \ln(t+1)^{1/4}} \Rightarrow \frac{1}{\ln e} \Rightarrow 1$$

Luego $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

3.- Calcular

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2} \right) =$$

4.- Calcular

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 6x} - (x-3)}{x+3 - \sqrt{x^2 + 6x}}$$

$$3) \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2} \right) \Rightarrow \lim_{x \rightarrow 0^+} \frac{x-1}{x^2} \Rightarrow -\infty$$

$$4) \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 6x} + x + 3}{x + 3 - \sqrt{x^2 + 6x}} \Rightarrow \lim_{x \rightarrow \infty} \frac{x\sqrt{1 - \frac{6}{x}} + x + 3}{x + 3 - x\sqrt{1 + \frac{6}{x}}} \Rightarrow \lim_{x \rightarrow \infty} \frac{x(\sqrt{1 - \frac{6}{x}} + 1) + 3}{x(1 - \sqrt{1 + \frac{6}{x}}) + 3} \Rightarrow \frac{x \cdot 2 + 3}{x \cdot 0 + 3} \Rightarrow \frac{\infty}{3} = \infty$$

$$\frac{\frac{1}{2\sqrt{x^2 - 6x}} \cdot 2x - 6 + 1}{1 - \frac{1}{2\sqrt{x^2 - 6x}} \cdot 2x + 6} \Rightarrow \frac{2x - 6 + 2\sqrt{x^2 - 6x}}{2\sqrt{x^2 - 6x}} \cdot \frac{2\sqrt{x^2 - 6x}}{2\sqrt{x^2 - 6x} - 2x + 6} \Rightarrow \frac{2\sqrt{x^2 - 6x} + 2x - 6}{2\sqrt{x^2 - 6x} - 2x + 6}$$

$$\Rightarrow \frac{2x\sqrt{1 - \frac{6}{x}} + 2x - 6}{2x\sqrt{1 - \frac{6}{x}} - 2x + 6} \Rightarrow \frac{2x(\sqrt{1 - \frac{6}{x}} + 1) - 6}{2x(\sqrt{1 - \frac{6}{x}} - 1) + 6} \Rightarrow \frac{2x \cdot 2 - 6}{2x \cdot 0 + 6} \Rightarrow \frac{\infty}{6} \Rightarrow \infty$$

Problemas de límites al infinito

Calcula los siguientes límites de funciones racionales, irracionales y exponenciales

$$a) \lim_{x \rightarrow \infty} \frac{2x^2 + x}{x^2 - 3} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{2x^2/x^2 + x/x^2}{x^2/x^2 - 3/x^2} = \lim_{x \rightarrow \infty} \frac{2 + 1/x}{1 - 3/x^2} = \frac{2}{1} = 2$$

$$b) \lim_{x \rightarrow \infty} \frac{x^3 + 5}{x^2 - x} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{x^3/x^3 + 5/x^3}{x^2/x^3 - x/x^3} = \lim_{x \rightarrow \infty} \frac{1 + 5/x^3}{1/x - 1/x^2} = \frac{1}{0} = \infty$$

$$c) \lim_{x \rightarrow \infty} \frac{x + 1}{x^4 + 2x + 2x} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{x/x^4 + 1/x^4}{x^4/x^4 + 2x/x^4 + \frac{2}{x^4}} =$$
$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x^3} + 1/x^4}{1 + 2/x^3 + \frac{2}{x^4}} = \frac{0}{1} = 0$$

$$d) \lim_{x \rightarrow \infty} \frac{5x^5 - 2}{2x^5 + x^3 - 1} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{5x^5/x^5 - 2/x^5}{2x^5/x^5 + x^3/x^5 - \frac{1}{x^5}} =$$
$$= \lim_{x \rightarrow \infty} \frac{5 - 2/x^5}{2 + 1/2 - \frac{1}{x^5}} = \frac{5}{2}$$

$$e) \lim_{x \rightarrow \infty} \frac{3x + 2}{x^2 + 7x + 1} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{3x/x^2 + 2/x^2}{x^2/x^2 + 7x/x^2 + \frac{1}{x^2}} =$$
$$= \lim_{x \rightarrow \infty} \frac{\frac{3}{x} + 2/x^2}{1 + 7/x + \frac{1}{x}} = \frac{0}{1} = 0$$

$$a) \lim_{x \rightarrow \infty} \frac{2x^2 + x}{x^2 - 3} :$$

$$\frac{\cancel{x^2} \left(2 + \frac{1}{\cancel{x}} \right)}{\cancel{x^2} \left(1 - \frac{3}{\cancel{x^2}} \right)} \Rightarrow 2$$

$$b) \lim_{x \rightarrow \infty} \frac{x^3 + 5}{x^2 - x} :$$

$$\frac{\cancel{x^3} \left(1 + \frac{5}{\cancel{x^3}} \right)}{\cancel{x^3} \left(\frac{1}{\cancel{x}} - \frac{x}{\cancel{x^2}} \right)} \Rightarrow \frac{1}{0} \quad \vee \quad \frac{\frac{x^3}{x^3} + \frac{5}{x^3}}{\frac{x^2}{x^3} - \frac{x}{x^3}} \Rightarrow \frac{1}{0}$$

$$c) \lim_{x \rightarrow \infty} \frac{x + 1}{x^4 + 2x + 2x}$$

$$\frac{\frac{x}{x^4} + \frac{1}{x^4}}{\frac{x^4}{x^4} + \frac{2x}{x^4} + \frac{2x}{x^4}} \Rightarrow 0$$

$$d) \lim_{x \rightarrow \infty} \frac{5x^5 - 2}{2x^5 + x^3 - 1} : \Rightarrow \frac{5}{2}$$

$$e) \lim_{x \rightarrow \infty} \frac{3x + 2}{x^2 + 7x + 1}$$

$$\frac{\frac{3x}{x^2} + \frac{2}{x^2}}{\frac{x^2}{x^2} + \frac{7x}{x^2} + \frac{1}{x^2}} \Rightarrow 0$$

$$\begin{aligned}
 f) \lim_{x \rightarrow \infty} \frac{2x^2 + x - 5}{x + 3x^2} &= \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{2x^2/x^2 + x/x^2 - 5/x^2}{x/x^2 + 3x^2/x^2} = \\
 &= \lim_{x \rightarrow \infty} \frac{2 + 1/x - 5/x^2}{1/x + 3} = \frac{2}{3}
 \end{aligned}$$

$$\begin{aligned}
 g) \lim_{x \rightarrow \infty} \frac{x^4 - 2}{\sqrt{(x^4 + 1)}} &= \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{x^4/x^4 - 2/x^4}{\sqrt{\left(\frac{x^4}{x^8} + \frac{1}{x^8}\right)}} = \lim_{x \rightarrow \infty} \frac{1 - 2/x^4}{\sqrt{\left(\frac{1}{x^4} + \frac{1}{x^8}\right)}} = \\
 &= \frac{1}{0} = \infty
 \end{aligned}$$

$$\begin{aligned}
 h) \lim_{x \rightarrow \infty} \frac{3x + \sqrt{x}}{1 - x^2} &= \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{\frac{3x}{x^2} + \sqrt{\frac{x}{x^4}}}{\frac{1}{x^2} - \frac{x^2}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{3}{x} + \sqrt{\frac{1}{x^3}}}{\frac{1}{x^2} - 1} \\
 &= \frac{0}{-1} = 0
 \end{aligned}$$

$$\begin{aligned}
 i) \lim_{x \rightarrow \infty} \frac{\sqrt{(4x^3 - 1)}}{x + 2} &= \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{\sqrt{\left(\frac{4x^3}{x^3} - \frac{1}{x^2}\right)}}{x/x + 2/x} = \\
 &= \lim_{x \rightarrow \infty} \frac{\sqrt{(4 - 1/x^2)}}{1 + 2/x} = \frac{2}{0} = \infty
 \end{aligned}$$

$$\begin{aligned}
 j) \lim_{x \rightarrow \infty} \frac{\sqrt{3x^3 + 2x - 1}}{\sqrt{x - 4}} &= \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{\sqrt{3x^3/x^3 + 2x/x^3 - 1/x^3}}{\sqrt{x/x^3 - 4/x^3}} = \\
 &= \lim_{x \rightarrow \infty} \frac{\sqrt{3 + 2/x^2 - 1/x^3}}{\sqrt{1/x^2 - 4/x^3}} = \frac{\sqrt{3}}{0} = \infty
 \end{aligned}$$

$$f) \lim_{x \rightarrow \infty} \frac{2x^2 + x - 5}{x + 3x^2} : \Rightarrow \frac{2}{3}$$

$$g) \lim_{x \rightarrow \infty} \frac{x^4 - 2}{\sqrt{(x^4 + 1)}}$$

$$\frac{\frac{x^4}{x^4} - \frac{2}{x^4}}{\frac{x^2}{x^4} \sqrt{\frac{1}{x^0} + \frac{1}{x^6}}} \Rightarrow \frac{1}{0} = \infty$$

$$h) \lim_{x \rightarrow \infty} \frac{3x + \sqrt{x}}{1 - x^2} =$$

$$\frac{\frac{3x}{x^2} + \sqrt{\frac{x}{x^4}}}{\frac{1}{x^2} - \frac{x^2}{x^2}} \Rightarrow \frac{0}{-1}$$

$$i) \lim_{x \rightarrow \infty} \frac{\sqrt{(4x^3 - 1)}}{x + 2} :$$

$$\sqrt{\frac{\frac{4x^3}{x^2} - \frac{1}{x^2}}{\frac{x}{x} + \frac{2}{x}}} \Rightarrow \frac{\sqrt{4x}}{1} \Rightarrow \infty$$

$$j) \lim_{x \rightarrow \infty} \frac{\sqrt{3x^3 + 2x - 1}}{\sqrt{x - 4}} :$$

$$\sqrt{\frac{\frac{3x^3}{x^0} + \frac{2x}{x^3} - \frac{1}{x^3}}{\sqrt{\frac{x}{x^0} - 4}}} \rightarrow \frac{\sqrt{3}}{0} = \infty$$

$$\begin{aligned}
k) \lim_{x \rightarrow \infty} (x - \sqrt{x^2 + x - 1}) &= \infty - \infty = \lim_{x \rightarrow \infty} \frac{(x - \sqrt{x^2 + x - 1}) \cdot (x + \sqrt{x^2 + x - 1})}{(x + \sqrt{x^2 + x - 1})} = \\
&= \lim_{x \rightarrow \infty} \frac{x^2 - (\sqrt{x^2 + x - 1})^2}{(x + \sqrt{x^2 + x - 1})} = \lim_{x \rightarrow \infty} \frac{x^2 - x^2 - x + 1}{x + \sqrt{x^2 + x - 1}} = \lim_{x \rightarrow \infty} \frac{-x/x + 1/x}{x/x + \sqrt{x^2/x^2 + x/x^2 - 1/x^2}} = \\
&= \lim_{x \rightarrow \infty} \frac{-1 + 1/x}{1 + \sqrt{1 + 1/x - 1/x^2}} = \frac{-1}{1 + \sqrt{1}} = -\frac{1}{2}
\end{aligned}$$

$$\begin{aligned}
l) \lim_{x \rightarrow \infty} \left(\frac{2x^3 - 1}{x + 2} - \frac{2x^3 + x + 1}{x} \right) &= \frac{\infty}{\infty} - \frac{\infty}{\infty} = \text{Doble indeterminacion} = \\
&= \lim_{x \rightarrow \infty} \frac{(2x^3 - 1) \cdot x - (2x^3 + x + 1) \cdot (x + 2)}{(x + 2) \cdot x} = \lim_{x \rightarrow \infty} \frac{2x^4 - x - 2x^4 - 4x^3 - x^2 - 2x - x - 2}{x^2 - 2x} = \\
&= \lim_{x \rightarrow \infty} \frac{-4x^3 - x^2 - 4x - 2}{x^2 - 2x} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{-4x^3/x^3 - x^2/x^3 - 4x/x^3 - 2/x^3}{x^2/x^3 - 2x/x^3} = \frac{-4}{0} = -\infty
\end{aligned}$$

$$\begin{aligned}
m) \lim_{x \rightarrow \infty} (\sqrt{2x^2 - 3} - \sqrt{2x^2 - 2x + 1}) &= \infty - \infty = \\
&= \lim_{x \rightarrow \infty} \frac{(\sqrt{2x^2 - 3} - \sqrt{2x^2 - 2x + 1}) \cdot (\sqrt{2x^2 - 3} + \sqrt{2x^2 - 2x + 1})}{\sqrt{2x^2 - 3} + \sqrt{2x^2 - 2x + 1}} = \\
&\lim_{x \rightarrow \infty} \frac{(\sqrt{2x^2 - 3})^2 - (\sqrt{2x^2 - 2x + 1})^2}{\sqrt{2x^2 - 3} + \sqrt{2x^2 - 2x + 1}} = \lim_{x \rightarrow \infty} \frac{2x^2 - 3 - 2x^2 + 2x - 1}{\sqrt{2x^2 - 3} + \sqrt{2x^2 - 2x + 1}} = \\
&= \lim_{x \rightarrow \infty} \frac{2x - 4}{\sqrt{2x^2 - 3} + \sqrt{2x^2 - 2x + 1}} = \frac{\infty}{\infty} = \\
&= \lim_{x \rightarrow \infty} \frac{2 - 4/x}{\sqrt{2 - 3/x^2} + \sqrt{2 - 2/x + 1/x^2}} = \frac{2}{\sqrt{2} + \sqrt{2}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}
\end{aligned}$$

$$k) \lim_{x \rightarrow \infty} (x - \sqrt{x^2 + x - 1})$$

$$x - \sqrt{x^2 + x - 1} \cdot \frac{x + \sqrt{x^2 + x - 1}}{x + \sqrt{x^2 + x - 1}} \Rightarrow \frac{x^2 - x^2 - x + 1}{x + \sqrt{x^2 + x - 1}} = \frac{-x + 1}{x + \sqrt{x^2 + x - 1}} \Rightarrow \frac{\frac{-x}{x} + \frac{1}{x}}{\frac{x}{x} + \sqrt{\frac{x^2}{x^2} + \frac{x}{x^2} - \frac{1}{x^2}}} \Rightarrow \frac{-1}{2}$$

$$\lim_{x \rightarrow \infty} \left(\frac{4x^3 - 1}{2x + 3} - \frac{2x^3 + 5}{x} \right) = \infty - \infty$$

$$= \lim_{x \rightarrow \infty} \frac{(4x^3 - 1) \cdot x - (2x^3 + 5) \cdot (2x + 3)}{(2x + 3) \cdot x}$$

$$= \lim_{x \rightarrow \infty} \frac{4x^4 - x - 4x^4 - 6x^3 - 10x - 15}{2x^2 + 3x} = \lim_{x \rightarrow \infty} \frac{-6x^3 - 11x - 15}{2x^2 + 3x} =$$

$$= \lim_{x \rightarrow \infty} \frac{-6 - 11/x^2 - 15/x^3}{2/x + 3/x^2} = \frac{-6}{0} = -\infty$$

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^3 - 2x^2 + 1} - \sqrt{x} \right) = \infty - \infty$$

$$= \lim_{x \rightarrow \infty} \frac{\left(\sqrt{x^3 - 2x^2 + 1} - \sqrt{x} \right) \left(\sqrt{x^3 - 2x^2 + 1} + \sqrt{x} \right)}{\sqrt{x^3 - 2x^2 + 1} + \sqrt{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{x^3 - 2x^2 + 1 - x}{\sqrt{x^3 - 2x^2 + 1} + \sqrt{x}} = \lim_{x \rightarrow \infty} \frac{\frac{x^3}{x^3} - \frac{2x^2}{x^3} - \frac{x}{x^3} + \frac{1}{x^3}}{\sqrt{\frac{x^3}{x^6} - \frac{2x^2}{x^6} + \frac{1}{x^6}} + \sqrt{\frac{x}{x^6}}}$$

$$= \lim_{x \rightarrow \infty} \frac{1 - \frac{2}{x} - \frac{1}{x^2} + \frac{1}{x^3}}{\sqrt{\frac{1}{x^3} - \frac{2}{x^4} + \frac{1}{x^6}} + \sqrt{\frac{1}{x^5}}} = \lim_{x \rightarrow \infty} \frac{1 - 0 - 0 + 0}{\sqrt{0 - 0 + 0} + \sqrt{0}}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{0} = \infty$$

$$\lim_{x \rightarrow \infty} \left(x - \sqrt{2x^2 + 1} \right) = \infty - \infty$$

$$= \lim_{x \rightarrow \infty} \frac{(x - \sqrt{2x^2 + 1})(x + \sqrt{2x^2 + 1})}{(x + \sqrt{2x^2 + 1})} = \lim_{x \rightarrow \infty} \frac{x^2 - 2x^2 - 1}{x + \sqrt{2x^2 + 1}}$$

$$= \lim_{x \rightarrow \infty} \frac{-x^2 - 1}{x + \sqrt{2x^2 + 1}} = \lim_{x \rightarrow \infty} \frac{\frac{-x^2}{x^2} - \frac{1}{x^2}}{\frac{x}{x^2} + \sqrt{\frac{2x^2}{x^4} + \frac{1}{x^4}}} = \lim_{x \rightarrow \infty} \frac{-1 - 0}{0 + \sqrt{0 + 0}}$$

$$\lim_{x \rightarrow \infty} \frac{-1}{0} = -\infty$$

$$\lim_{x \rightarrow \infty} \left(\frac{2x^3 - 2}{x + 3} - \frac{x^2 + 1}{x} \right) = [\infty - \infty]$$

$$= \lim_{x \rightarrow \infty} \frac{(2x^3 - 2) \cdot x - (x^2 + 1)(x + 3)}{x(x + 3)}$$

$$= \lim_{x \rightarrow \infty} \frac{2x^4 - 2x - x^3 - 3x^2 - x - 3}{x^2 + 3x}$$

$$= \lim_{x \rightarrow \infty} \frac{2x^4 - 3x - x^3 - 3x^2 - 3}{x^2 + 3x} = \frac{2}{0} = \infty$$

$$\begin{aligned}
& \lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 1} - \sqrt{x^2 + 1} \right) = \infty - \infty = \\
& = \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 - 1} - \sqrt{x^2 + 1})(\sqrt{x^2 - 1} + \sqrt{x^2 + 1})}{\sqrt{x^2 - 1} + \sqrt{x^2 + 1}} = \\
& = \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 - 1})^2 - (\sqrt{x^2 + 1})^2}{\sqrt{x^2 - 1} + \sqrt{x^2 + 1}} = \lim_{x \rightarrow \infty} \frac{x^2 - 1 - x^2 - 1}{\sqrt{x^2 - 1} + \sqrt{x^2 + 1}} = \\
& = \lim_{x \rightarrow \infty} \frac{-2}{\sqrt{x^2 - 1} + \sqrt{x^2 + 1}} = \frac{-2}{\infty} = 0
\end{aligned}$$

$$\begin{aligned}
& \lim_{x \rightarrow \infty} \left(\frac{x^3 - 2x}{x^2 - 1} - \frac{x^2 + 3}{x + 1} \right) = \infty - \infty = \\
& = \lim_{x \rightarrow \infty} \frac{(x^3 - 2x) - (x^2 + 3)(x + 1)}{x^2 - 1} = \\
& = \lim_{x \rightarrow \infty} \frac{x^3 - 2x - x^3 - x^2 - 3x - 3}{x^2 - 1} = \lim_{x \rightarrow \infty} \frac{-x^2 - 5x - 3}{x^2 - 1} = 1
\end{aligned}$$

$$\lim_{x \rightarrow \infty} (\sqrt{2x^2 + 1} - \sqrt{x^2 - 3}) = [\infty - \infty] =$$

$$\begin{aligned}
& \lim_{x \rightarrow \infty} \frac{(\sqrt{2x^2 + 1} - \sqrt{x^2 - 3})(\sqrt{2x^2 + 1} + \sqrt{x^2 - 3})}{\sqrt{2x^2 + 1} + \sqrt{x^2 - 3}} = \\
& = \lim_{x \rightarrow \infty} \frac{(\sqrt{2x^2 + 1})^2 - (\sqrt{x^2 - 3})^2}{\sqrt{2x^2 + 1} + \sqrt{x^2 - 3}} = \\
& = \lim_{x \rightarrow \infty} \frac{2x^2 + 1 - x^2 + 3}{\sqrt{2x^2 + 1} + \sqrt{x^2 - 3}} = \lim_{x \rightarrow \infty} \frac{x^2 + 4}{\sqrt{2x^2 + 1} + \sqrt{x^2 - 3}} = \infty
\end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \left(\frac{2x-1}{x^2+3} \times \frac{2x^3-2}{x} \right) &= 0 \times \infty \\
 &= \lim_{x \rightarrow \infty} \frac{4x^4 - 4x - 2x^3 + 2}{x^3 + 3x} = \lim_{x \rightarrow \infty} \frac{\frac{4x^4}{x^4} - \frac{4x}{x^4} - \frac{2x^3}{x^4} + \frac{2}{x^4}}{\frac{x^3}{x^4} + \frac{3x}{x^4}} \\
 \lim_{x \rightarrow \infty} \frac{4-0-0+0}{0+0} &= \lim_{x \rightarrow \infty} \frac{4}{0} = \infty
 \end{aligned}$$

$$\lim_{x \rightarrow \infty} \left(\frac{4x^2}{x^3+2} \cdot \frac{3x^2-1}{x+3} \right) = [0 \text{ } \infty]$$

$$= \lim_{x \rightarrow \infty} \frac{12x^4 - 4x^2 - 3x^2 + 1}{x^4 - 3x^3 - 2x + 6} = \lim_{x \rightarrow \infty} \frac{12x^4 - 7x^2 + 1}{x^4 - 3x^3 - 2x + 6} = 12$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{2x-1} - \sqrt{2x+1}}{x} = \frac{[\infty - \infty]}{x} =$$

$$\lim_{x \rightarrow \infty} \frac{(\sqrt{2x-1} - \sqrt{2x+1})(\sqrt{2x-1} + \sqrt{2x+1})}{x(\sqrt{2x-1} + \sqrt{2x+1})} = \lim_{x \rightarrow \infty} \frac{(\sqrt{2x-1})^2 - (\sqrt{2x+1})^2}{x\sqrt{2x-1} + x\sqrt{2x+1}} =$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{2x} - 1 - \cancel{2x} - 1}{x\sqrt{2x-1} + x\sqrt{2x+1}} = \lim_{x \rightarrow \infty} \frac{-2}{x\sqrt{2x-1} + x\sqrt{2x+1}} = 0$$

$$\lim_{x \rightarrow \infty} \left(\frac{2x^3 - x + 1}{2x^3 + 3x} \right)^{x^2-1} = [1^\infty] = e^p$$

$$\begin{aligned}
 p &= \lim_{x \rightarrow \infty} (x^2 - 1) \cdot \left[\frac{2x^3 - x + 1}{2x^3 + 3x} - 1 \right] = \lim_{x \rightarrow \infty} (x^2 - 1) \cdot \left(\frac{2x^3 - x + 1 - 2x^3 - 3x}{2x^3 + 3x} \right) = \\
 &= \lim_{x \rightarrow \infty} \frac{(x^2 - 1)(1 - 4x)}{2x^3 + 3x} = \lim_{x \rightarrow \infty} \frac{x^2 - 4x^3 - 1 + 4x}{2x^3 + 3x} = \frac{-4}{2} = -2 \\
 \lim_{x \rightarrow \infty} \left(\frac{2x^3 - x + 1}{2x^3 + 3x} \right)^{x^2-1} &= e^p = e^{-2} = \frac{1}{e^2}
 \end{aligned}$$

$$\lim_{x \rightarrow \infty} \left(\frac{4x^2 - 3}{4x^2 + 3} \right)^{2x^2 - 1} = [1^\infty] = e^p$$

$$p = \lim_{x \rightarrow \infty} (2x^2 - 1) \cdot \left[\frac{4x^2 - 3}{4x^2 + 3} - 1 \right] = \lim_{x \rightarrow \infty} (2x^2 - 1) \cdot \left(\frac{4x^2 - 3 - 4x^2 - 3}{4x^2 + 3} \right) = \lim_{x \rightarrow \infty} \frac{(2x^2 - 1)(-6)}{4x^2 + 3} =$$

$$\lim_{x \rightarrow \infty} \frac{-12x^2 + 6}{4x^2 + 3} = -3$$

$$\lim_{x \rightarrow \infty} \left(\frac{4x^2 - 3}{4x^2 + 3} \right)^{x^2 - 1} = e^p = e^{-3} = \frac{1}{e^3}$$

$$\lim_{x \rightarrow \infty} \left(\frac{x^2 + 3}{x^2 - x} \right)^{4x + 1} = 1^\infty = e^p$$

$$p = \lim_{x \rightarrow \infty} (4x + 1) \left[\frac{x^2 + 3}{x^2 - x} - 1 \right] = \lim_{x \rightarrow \infty} (4x + 1) \cdot \frac{x^2 + 3 - x^2 + x}{x^2 - x} =$$

$$\lim_{x \rightarrow \infty} \frac{(4x + 1) \cdot (3 + x)}{x^2 - x} = \lim_{x \rightarrow \infty} \frac{4x^2 + 13x + 3}{x^2 - x} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{4 + 13/x + 3/x^2}{1 - 1/x} = 4$$

$$\lim_{x \rightarrow \infty} \left(\frac{x^2 + 3}{x^2 - x} \right)^{4x + 1} = e^p = e^4$$

$$\lim_{x \rightarrow \infty} \left(\frac{x^3 + 2x^2}{x^3 - 1} \right)^{3x^2 + 1} = 1^\infty = e^p$$

$$p = \lim_{x \rightarrow \infty} (3x^2 + 1) \cdot \left(\frac{x^3 + 2x^2}{x^3 - 1} - 1 \right) = \lim_{x \rightarrow \infty} (3x^2 + 1) \cdot \frac{x^3 + 2x^2 - x^3 + 1}{x^3 - 1} =$$

$$= \lim_{x \rightarrow \infty} \frac{(3x^2 + 1) \cdot (2x^2 + 1)}{x^3 - 1} = \lim_{x \rightarrow \infty} \frac{6x^4 + 5x^2 + 1}{x^3 - 1} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{6 + 5/x^2 + 1/x^4}{1/x - 1/x^4} =$$

$$\frac{6}{0} = \infty$$

$$\lim_{x \rightarrow \infty} \left(\frac{x^3 + 2x^2}{x^3 - 1} \right)^{3x^2 + 1} = e^p = e^\infty = \infty$$

$$\lim_{x \rightarrow \infty} \left(\frac{3x + 1}{3x} \right)^{x^2} = 1^\infty = e^p$$

$$p = \lim_{x \rightarrow \infty} x^2 \cdot \left(\frac{3x + 1}{3x} - 1 \right) = \lim_{x \rightarrow \infty} x^2 \cdot \left(\frac{3x + 1 - 3x}{3x} \right) = \lim_{x \rightarrow \infty} \frac{x^2}{3x} = \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{3/x} = \frac{1}{0} = \infty$$

$$\lim_{x \rightarrow \infty} \left(\frac{3x + 1}{3x} \right)^{x^2} = e^\infty = \infty$$

Calcular los siguientes límites:

$$\text{a) } \lim_{x \rightarrow \infty} \frac{x + \sqrt{x}}{\sqrt{x^2 + 1}} = \frac{\infty}{\sqrt{\infty}} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{\frac{x + \sqrt{x}}{x}}{\frac{\sqrt{x^2 + 1}}{x}} = \lim_{x \rightarrow \infty} \frac{1 + \sqrt{1/x}}{\sqrt{1 + 1/x^2}} = \frac{1 + 0}{\sqrt{1 + 0}} = \frac{1}{1} = 1$$

↑

Dividimos numerador y denominador por la mayor potencia de x es decir por x.

$$\text{b) } \lim_{x \rightarrow 2} \frac{\sqrt{x^3 - 8}}{x^2 - 4} = \frac{\sqrt{8 - 8}}{4 - 4} = \left[\frac{0}{0} \right] = \lim_{x \rightarrow 2} \frac{\sqrt{(x-2)(x^2 + 2x + 4)}}{(x+2)(x-2)} ;$$

Descomponemos en factores el radicando y el denominador y luego simplificamos términos

$$x^3 - 8 = 0$$

$$\begin{array}{r|rrrr} & 1 & 0 & 0 & -8 \\ 2 & & 2 & 4 & 8 \\ \hline & 1 & 2 & 4 & 0 \end{array}$$

$$x^2 + 2x + 4 = 0 ; x = \frac{-2 \pm \sqrt{4 - 16}}{2} \quad \text{no se puede resolver}$$

$$x^2 - 4 = 0 ; x = \pm 2 ; x^2 - 4 = (x+2)(x-2)$$

$$\lim_{x \rightarrow 2} \frac{\sqrt{x^3 - 8}}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{(x-2)^{1/2} \sqrt{x^2 + 2x + 4}}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 2x + 4}}{(x-2)^{1/2} (x+2)} = \frac{\sqrt{4 + 4 + 4}}{0 \cdot 4} =$$

$$= \frac{\sqrt{12}}{0} = \infty$$

$$\text{c) } \lim_{x \rightarrow \infty} (\sqrt{2x^3 - 1} - \sqrt{x^3}) = \sqrt{\infty} - \sqrt{\infty} = [\infty - \infty] = \lim_{x \rightarrow \infty} \frac{(\sqrt{2x^3 - 1} - \sqrt{x^3})(\sqrt{2x^3 - 1} + \sqrt{x^3})}{\sqrt{2x^3 - 1} + \sqrt{x^3}}$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{2x^3 - 1})^2 - (\sqrt{x^3})^2}{\sqrt{2x^3 - 1} + \sqrt{x^3}} = \lim_{x \rightarrow \infty} \frac{2x^3 - 1 - x^3}{\sqrt{2x^3 - 1} + \sqrt{x^3}} = \lim_{x \rightarrow \infty} \frac{x^3 - 1}{\sqrt{2x^3 - 1} + \sqrt{x^3}} = \left[\frac{\infty}{\infty} \right] =$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^3 - 1}{x^3}}{\frac{\sqrt{2x^3 - 1} + \sqrt{x^3}}{x^3}} = \lim_{x \rightarrow \infty} \frac{1 - 1/x^3}{\sqrt{2/x^3 - 1/x^6} + \sqrt{1/x^3}} = \frac{1}{0} = \infty$$

$$\text{d) } \lim_{x \rightarrow \infty} \frac{8x^2 + 3x + 2}{\sqrt{x^3 + x^4}} = \frac{\infty}{\sqrt{\infty}} = \left[\frac{\infty}{\infty} \right] = \lim_{x \rightarrow \infty} \frac{\frac{8x^2 + 3x + 2}{x^2}}{\sqrt{\frac{x^3 + x^4}{x^4}}} = \lim_{x \rightarrow \infty} \frac{8 + \frac{3}{x} + \frac{2}{x^2}}{\sqrt{\frac{1}{x} + 1}} = \frac{8}{\sqrt{1}} = 8$$

Calcular los siguientes límites:

$$\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 4x - 1}) = [\infty - \infty] = \lim_{x \rightarrow \infty} \frac{(x - \sqrt{x^2 + 4x - 1}) \cdot (x + \sqrt{x^2 + 4x - 1})}{(x + \sqrt{x^2 + 4x - 1})} =$$

$$\lim_{x \rightarrow \infty} \frac{x^2 - (\sqrt{x^2 + 4x - 1})^2}{x + \sqrt{x^2 + 4x - 1}} = \lim_{x \rightarrow \infty} \frac{x^2 - x^2 - 4x - 1}{x + \sqrt{x^2 + 4x - 1}} = \lim_{x \rightarrow \infty} \frac{4x + 1}{x + \sqrt{x^2 + 4x - 1}} = \left[\frac{-\infty}{\infty} \right] =$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{4x + 1}{x}}{\frac{x + \sqrt{x^2 + 4x - 1}}{x}} = \lim_{x \rightarrow \infty} \frac{4x + \frac{1}{x}}{1 + \sqrt{1 + \frac{4}{x} - \frac{1}{x^2}}} = \frac{-4}{1 + 1} = -2$$

$$\lim_{x \rightarrow \infty} (\sqrt{4x^2 + x + 31} - 3x) = [\infty - \infty] = \lim_{x \rightarrow \infty} \frac{(\sqrt{4x^2 + x + 31} - 3x) \cdot (\sqrt{4x^2 + x + 31} + 3x)}{\sqrt{4x^2 + x + 31} + 3x} =$$

$$\lim_{x \rightarrow \infty} \frac{(\sqrt{4x^2 + x + 31})^2 - (3x)^2}{\sqrt{4x^2 + x + 31} + 3x} = \lim_{x \rightarrow \infty} \frac{4x^2 + x + 31 - 9x^2}{\sqrt{4x^2 + x + 31} + 3x} =$$

$$\lim_{x \rightarrow \infty} \frac{-5x^2 + x + 31}{\sqrt{4x^2 + x + 31} + 3x} = \left[\frac{-\infty}{\infty} \right] = \lim_{x \rightarrow \infty} \frac{5 - \frac{1}{x} + \frac{31}{x^2}}{\sqrt{\frac{4}{x^2} + \frac{1}{x^3} + \frac{31}{x^4}} + \frac{3}{x}} = \frac{-5}{0} = -\infty$$

$$\lim_{x \rightarrow \infty} (4x - \sqrt{16x^2 + 2}) = [\infty - \infty] = \lim_{x \rightarrow \infty} \frac{(4x - \sqrt{16x^2 + 2})(4x + \sqrt{16x^2 + 2})}{(4x + \sqrt{16x^2 + 2})} =$$

$$\lim_{x \rightarrow \infty} \frac{(4x)^2 - (\sqrt{16x^2 + 2})^2}{4x + \sqrt{16x^2 + 2}} = \lim_{x \rightarrow \infty} \frac{16x^2 - 16x^2 - 2}{4x + \sqrt{16x^2 + 2}} =$$

$$\lim_{x \rightarrow \infty} \frac{-2}{4x + \sqrt{16x^2 + 2}} = \frac{-2}{\infty + \infty} = \frac{-2}{\infty} = \mathbf{0}$$

$$\lim_{x \rightarrow 1} \frac{-x+1}{\sqrt{x+4}-\sqrt{5}} = \frac{-1+1}{\sqrt{5}-\sqrt{5}} = \frac{0}{0} = \lim_{x \rightarrow 1} \frac{(-x+1)(\sqrt{x+4}+\sqrt{5})}{(\sqrt{x+4}-\sqrt{5})(\sqrt{x+4}+\sqrt{5})} =$$

$$\lim_{x \rightarrow 1} \frac{(-x+1)(\sqrt{x+4}+\sqrt{5})}{x+4-5} = \lim_{x \rightarrow 1} \frac{(-x+1)(\sqrt{x+4}+\sqrt{5})}{x-1} =$$

$$\lim_{x \rightarrow 1} \frac{-(x+1)(\sqrt{x+4}+\sqrt{5})}{x-1} = -(\sqrt{5}+\sqrt{5}) = \mathbf{-2\sqrt{5}}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^3-1}}{x^2+3} &= \frac{\sqrt[3]{\infty}}{\infty} = \left[\frac{\infty}{\infty} \right] = \lim_{x \rightarrow \infty} \frac{\frac{\sqrt[3]{x^3-1}}{x^2}}{x^2+3} = \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{(x^3-1)}{x^4}}}{\frac{x^2+3}{x^2}} = \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{1/x - 1/x^4}}{1 + 3/x^2} = \frac{\sqrt{0}}{1} = \frac{0}{1} = \mathbf{0} \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow 3} \frac{\sqrt{x+6} - 3}{x - 3} &= \left[\frac{0}{0} \right] = \lim_{x \rightarrow 3} \frac{(\sqrt{x+6} - 3) \cdot (\sqrt{x+6} + 3)}{(x-3) \cdot (\sqrt{x+6} + 3)} = \\
 &= \lim_{x \rightarrow 3} \frac{(\sqrt{x+6})^2 - 3^2}{(x-3) \cdot (\sqrt{x+6} + 3)} = \lim_{x \rightarrow 3} \frac{x + 6 - 9}{(x-3) \cdot (\sqrt{x+6} + 3)} = \\
 &= \lim_{x \rightarrow 3} \frac{(\cancel{x-3})}{(\cancel{x-3}) \cdot (\sqrt{x+6} + 3)} = \lim_{x \rightarrow 3} \frac{1}{(\sqrt{x+6} + 3)} = \frac{1}{3 + 3} = \frac{1}{6}
 \end{aligned}$$

$$\lim_{x \rightarrow 1/3} \frac{3x^3 - 7x^2 + 5x - 1}{3x^3 + 2x^2 - 7x + 2} = \frac{3 \cdot 1/27 - 7 \cdot 1/9 + 5 \cdot 1/3 - 1}{3 \cdot 1/27 + 2 \cdot 1/9 - 7 \cdot 1/3 + 2} = \left[\frac{0}{0} \right]$$

• Factorización
 $3x^3 - 7x^2 + 5x - 1 = 0$

$$\begin{array}{r|rrrr}
 & 3 & -7 & 5 & -1 \\
 1/3 & & 1 & -2 & 1 \\
 \hline
 & 3 & -6 & 3 & 0
 \end{array} \rightarrow 3x^2 - 6x + 3 = 0; \quad x = \frac{6 \pm \sqrt{36 - 36}}{6} = \begin{matrix} 1 \\ 1 \end{matrix}$$

$$3x^3 - 7x^2 + 5x - 1 = 3(x-1)^2 \cdot (x-1/3)$$

$$\underline{3x^3 + 2x^2 - 7x + 2 = 0}$$

$$\begin{array}{r|rrrr}
 & 3 & 2 & -7 & 2 \\
 1/3 & & 1 & 1 & -2 \\
 \hline
 & 3 & 3 & -6 & 0
 \end{array} \rightarrow 3x^2 + 3x - 6 = 0; \quad x = \frac{-3 \pm \sqrt{9+72}}{6} = \begin{matrix} 1 \\ -2 \end{matrix}$$

$$3x^3 + 2x^2 - 7x + 2 = 3(x-1) \cdot (x+2) \cdot (x-1/3)$$

$$\lim_{x \rightarrow 1/3} \frac{3(x-1)^2 \cdot (\cancel{x-1/3})}{3(\cancel{x-1}) \cdot (x+2) \cdot (\cancel{x-1/3})} = \lim_{x \rightarrow 1/3} \frac{(x-1)}{(x+2)} = \frac{1/3 - 1}{1/3 + 2} = \frac{-2/3}{7/3} = \frac{-2}{7}$$

$$d) \lim (\sqrt{x+3} - \sqrt{x-3}) = \sqrt{\infty} - \sqrt{\infty} = [\infty - \infty] =$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{x+3} - \sqrt{x-3}) \cdot (\sqrt{x+3} + \sqrt{x-3})}{\sqrt{x+3} + \sqrt{x-3}} = \lim_{x \rightarrow \infty} \frac{(\sqrt{x+3})^2 - (\sqrt{x-3})^2}{\sqrt{x+3} + \sqrt{x-3}} =$$

$$= \lim_{x \rightarrow \infty} \frac{x+3 - (x-3)}{\sqrt{x+3} + \sqrt{x-3}} = \lim_{x \rightarrow \infty} \frac{6}{\sqrt{x+3} + \sqrt{x-3}} = \frac{6}{\infty + \infty} = \frac{6}{\infty} = 0 /$$

Calcular los siguientes limites

1)

$$\lim_{x \rightarrow \infty} \left(\frac{5x^2+x}{x-3} - \frac{x^2+2}{x-1} \right) = [\infty - \infty]$$

$$\lim_{x \rightarrow \infty} \frac{(5x^2+x) \cdot (x-1) - (x^2+2) \cdot (x-3)}{(x-3) \cdot (x-1)} = \lim_{x \rightarrow \infty} \frac{5x^3 - 5x^2 + x^2 - x - x^3 + 3x^2 - 2x + 6}{x^2 - x - 3x + 1} =$$

$$= \lim_{x \rightarrow \infty} \frac{4x^3 - x^2 - 3x + 6}{x^2 - 4x + 1} = \left[\frac{\infty}{\infty} \right] = \lim_{x \rightarrow \infty} \frac{4 - \frac{1}{x} - \frac{3}{x^2} + \frac{6}{x^3}}{\frac{1}{x} - \frac{4}{x^2} + \frac{1}{x^3}} = \frac{4}{0} = \infty]$$

2)

$$\lim_{x \rightarrow \infty} \frac{(x^2+x) \cdot (x-1) - (x^2+2) \cdot (x-3)}{(x-3) \cdot (x-1)} = [\infty - \infty] = \lim_{x \rightarrow \infty} \frac{(x^2-3x+2) \cdot (x+3) - x^2 \cdot 3x}{3x(x+3)} =$$

$$= \lim_{x \rightarrow \infty} \frac{x^3 + 3x^2 - 3x^2 - 9x + 2x + 6 - 3x^3}{3x^2 + 3x} = \lim_{x \rightarrow \infty} \frac{-2x^3 - 7x + 6}{3x^2 + 3x} = \left[\frac{\infty}{\infty} \right] =$$

$$= \lim_{x \rightarrow \infty} \frac{-2 - \frac{7}{x^2} + \frac{6}{x^3}}{\frac{3}{x} + \frac{3}{x^2}} = \frac{-2}{0} = -\infty]$$

3)

$$\lim_{x \rightarrow \infty} \left(2x - \frac{4x^2-2x+7}{2x+1} \right) = \lim_{x \rightarrow \infty} \frac{2x(2x+1) - (4x^2-2x+7)}{2x+1} = \lim_{x \rightarrow \infty} \frac{4x^2+2x-4x^2+2x-7}{2x+1} =$$

$$= \lim_{x \rightarrow \infty} \frac{4x-7}{2x+1} = \left[\frac{\infty}{\infty} \right] = \lim_{x \rightarrow \infty} \frac{4 - \frac{7}{x}}{2 + \frac{1}{x}} = \frac{4}{2} = 2]$$

4)

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(\frac{6x^2+1}{2x+4} - \frac{9x^2-5}{3x+6} \right) &= \lim_{x \rightarrow \infty} \frac{(6x^2+1)-(3x+6)-(9x^2-5) \cdot (2x+4)}{(2x+4) \cdot (3x+6)} = \\ &= \lim_{x \rightarrow \infty} \frac{18x^2+36x^2+3x+6-18x^3-36x^2+10x+20}{6x^2+12x+12x+24} = \lim_{x \rightarrow \infty} \frac{13x+26}{6x^2+24x+24} = \\ &= \left[\frac{\infty}{\infty} \right] = \lim_{x \rightarrow \infty} \frac{\frac{13}{x} + \frac{26}{x}}{6 + \frac{24}{x} + \frac{24}{x^2}} = \frac{0}{6} = 0 \end{aligned}$$

LIMITES LATERALES , INFINITOS Y EN EL INFINITO, ASÍNTOTAS

Listado 3- Límites laterales, infinitos y en el infinito, asíntotas

1. En cada caso determine si el límite indicado existe e indique su valor.

$$a) \ f(x) = \begin{cases} x^2; & x \leq -1 \\ x^3; & x > -1 \end{cases}$$

$$\lim_{x \rightarrow -1} f(x)$$

$$d) \ f(x) = \begin{cases} 9 - x^2; & x > 3 \\ 3; & x = 3 \\ x^2 - 9; & x < 3 \end{cases}$$

$$\lim_{x \rightarrow 3} f(x)$$

$$b) \ g(x) = \begin{cases} -2x; & x \leq 2 \\ x^2 - 4x; & x > 2 \end{cases}$$

$$\lim_{x \rightarrow 2} g(x)$$

$$e) \ g(x) = \begin{cases} x + 5; & x \leq -3 \\ \sqrt{9 - x^2}; & -3 < x < 3 \\ 3 - x; & x \geq 3 \end{cases}$$

$$\lim_{x \rightarrow 3} g(x), \lim_{x \rightarrow -3} g(x), \lim_{x \rightarrow 0} g(x)$$

$$c) \ h(x) = \begin{cases} \frac{x^2 + 3x - 10}{x + 5}; & x > -5 \\ -3; & x \leq -5 \end{cases}$$

$$\lim_{x \rightarrow -5} h(x)$$

$$f) \ f(x) = \begin{cases} |x - 1|; & x < -1 \\ 0; & x = -1 \\ |1 - x|; & x > -1 \end{cases}$$

$$\lim_{x \rightarrow 1} f(x), \lim_{x \rightarrow -1} f(x)$$

2. Sean f y g las funciones definidas como

$$f(x) = \begin{cases} x + 1; & x < 1 \\ x - 1; & x \geq 1 \end{cases} \quad g(x) = \begin{cases} 1 - x; & x < 1 \\ 1 + x; & x \geq 1 \end{cases}$$

a) Muestre que $\lim_{x \rightarrow 1} f(x)$ y $\lim_{x \rightarrow 1} g(x)$ no existen

b) Defina la función $f + g$

c) Muestre que $\lim_{x \rightarrow 1} [f(x) + g(x)]$ existe

d) De los resultados de a) y c) se tiene que

$$\lim_{x \rightarrow 1} f(x) + \lim_{x \rightarrow 1} g(x) \neq \lim_{x \rightarrow 1} [f(x) + g(x)]$$

¿Contradice este hecho el teorema correspondiente del álgebra de límites?

3. Calcule los siguientes límites si existen, o indique su tendencia.

$$\begin{array}{ll}
a) \lim_{x \rightarrow 5^-} \frac{\sqrt{5-x^2}}{x-5} = \infty & \tilde{n}) \lim_{x \rightarrow 1} \frac{2x^2}{1-x^2} = \\
b) \lim_{x \rightarrow 3^+} \frac{\sqrt{x^2-6x+9}}{x-3} = 1 & o) \lim_{x \rightarrow -1^-} \frac{2x}{x-1} = \\
c) \lim_{x \rightarrow 0^-} \frac{x}{x-|x|} = \frac{\infty}{\infty} & p) \lim_{x \rightarrow 2^-} \frac{(x-3)\sqrt{4-x^2}}{x^2-4} = \\
d) \lim_{x \rightarrow 2} 3+|2x-4| = & q) \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2}\right) = \\
e) \lim_{x \rightarrow 3^+} \frac{2-x}{(x-3)^4} = & r) \lim_{x \rightarrow 0^-} \frac{x^2-3}{x^3+x^2} = \\
f) \lim_{x \rightarrow +\infty} \frac{4x^2+2x+1}{2x+5} = & s) \lim_{x \rightarrow -\infty} \frac{x^2-3}{x^3+x^2} = \\
g) \lim_{x \rightarrow -\infty} \frac{3x^2-3x-1}{\sqrt{x^4+1}} = & t) \lim_{x \rightarrow -\infty} \frac{3x+4}{\sqrt{2x^2-5}} = \\
h) \lim_{x \rightarrow +\infty} \frac{2x^2-x+5}{4x^3-1} = & u) \lim_{x \rightarrow +\infty} \frac{2x-x^2}{3x+5} = \\
i) \lim_{x \rightarrow 0^-} \frac{|x|-2}{x-|x|} = & v) \lim_{x \rightarrow +\infty} \sqrt{3x^2+x}-2x = \\
j) \lim_{x \rightarrow +\infty} \frac{3x+4}{x+\sqrt{2x^2-5}} = & w) \lim_{x \rightarrow +\infty} \frac{\sqrt{x+\sqrt{x+\sqrt{x}}}}{\sqrt{x+1}} = \\
k) \lim_{x \rightarrow -\infty} \frac{3x+4}{x+\sqrt{2x^2-5}} = & x) \lim_{x \rightarrow \frac{\pi}{2}^-} \tan x = \infty \\
l) \lim_{x \rightarrow +\infty} \sqrt{x^2+x}-\sqrt{x^2-5} = & y) \lim_{x \rightarrow +\infty} \frac{\sin 2x}{x} = 0 \\
m) \lim_{x \rightarrow -\infty} \sqrt{x^2+x}-\sqrt{x^2-5} = & z) \lim_{x \rightarrow +\infty} \frac{1}{2x+\sin x} = \frac{1}{\infty} \cdot \frac{1}{\frac{2x+\sin x}{x}} \Rightarrow \frac{1}{3x} \Rightarrow 0 \\
n) \lim_{x \rightarrow -\infty} \frac{3x^3+4}{x+\sqrt{2x^2-5}} = &
\end{array}$$

4. Determine las asíntotas y esboze la gráfica de la función f definida por:

$$\begin{array}{ll}
a) f(x) = \frac{x}{x+3} & A. = -3 \quad \text{[Hand-drawn graph showing a horizontal asymptote at } y = -3 \text{ and a vertical asymptote at } x = -3 \text{]} \\
b) f(x) = \frac{x^2}{x+3} & A = -3 \\
c) f(x) = \frac{1}{(x-2)^2} & A = 2 \\
d) f(x) = \frac{x^3}{x^2-1} & A = 1 \wedge -1 \\
e) f(x) = \frac{3x^2-2}{x^2-x-2} & A = 2 \wedge -1 \\
f) f(x) = \frac{2(x^2-9)}{x^2-4} & A = -2 \wedge 2 \\
g) f(x) = \frac{-x^3+x^2+4}{x^2} & A = 0 \\
h) f(x) = \frac{x}{\sqrt{x^2+2}} & A = X \\
i) f(x) = \tan x &
\end{array}$$

5. Calcular los siguientes límites trigonométricos:

$$(a) \lim_{x \rightarrow 0} \frac{\sin(2x)}{x}$$

$$(b) \lim_{x \rightarrow 0} \frac{2 - 2\cos(x)}{2x}$$

$$(c) \lim_{x \rightarrow 0} x^3 \cos\left(\frac{1}{x^4}\right)$$

$$(d) \lim_{x \rightarrow 1} \frac{\sin(\pi x)}{x - 1}$$

$$(e) \lim_{x \rightarrow 0} \frac{\sin(x^2) \tan(2x)}{x \sin^2(3x)}$$

$$(f) \lim_{x \rightarrow 1} \frac{\sin(x^2 - 1)}{x - 1}$$

$$(g) \lim_{x \rightarrow \pi} \frac{\sin(7x)}{\sin(5x)}$$

$$(h) \lim_{x \rightarrow \pi} \frac{\tan(2x)}{x - \pi}$$

$$(i) \lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin(x) - \sin\left(\frac{\pi}{6}\right)}{x - \frac{\pi}{6}}$$

$$(j) \lim_{x \rightarrow 0} \frac{\sin(3\pi x)}{\sin(8\pi x)}$$

$$(k) \lim_{x \rightarrow 0} \frac{\frac{1 - \cos(x)}{x} - \frac{1}{4}x}{x}$$

$$(l) \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin(3x)}{x(3x - \pi)}$$

$$a) \lim_{x \rightarrow 0} \frac{\sin(2x)}{x} \Rightarrow \lim_{x \rightarrow 0} \frac{\sin(2x)}{x} \cdot \frac{2}{2} \Rightarrow \lim_{x \rightarrow 0} \frac{2\sin(2x)}{2x} = 2$$

$$b) \lim_{x \rightarrow 0} \frac{2 - 2\cos(x)}{2x} \Rightarrow \frac{2(1 - \cos x)}{2x} = 0$$

$$c) \lim_{x \rightarrow 0} x^3 \cos\left(\frac{1}{x^4}\right) \rightarrow \text{Por teorema del acotamiento tenemos algo nulo Por algo acotado}$$

$$d) \lim_{x \rightarrow 1} \frac{\sin(\pi x)}{x - 1} \Rightarrow \begin{matrix} u = x - 1 \\ x = u + 1 \\ u \rightarrow 0 \end{matrix} \Rightarrow \lim_{u \rightarrow 0} \frac{\sin(\pi u + \pi)}{u} \Rightarrow \lim_{u \rightarrow 0} \frac{\sin(\pi u + \pi)}{u}$$

$$\lim_{u \rightarrow 0} \frac{\sin(\pi u) \overset{1}{\cancel{\cos(\pi)}} + \cos(\pi u) \overset{0}{\cancel{\sin(\pi)}}}{u} \Rightarrow \frac{\sin(u\pi)}{u} \cdot \frac{\pi}{\pi} = \pi$$

$$e) \lim_{x \rightarrow 0} \frac{\sin^2(x) \tan(2x)}{x \sin^2(3x)} = \frac{2\sin^2(x)}{\sin^2(3x)} \cdot \frac{x}{x} \cdot \frac{3}{3} \Rightarrow \frac{\cancel{3x} \cdot 2 \cdot \cancel{\sin x} \cdot \cancel{\sin x}}{\cancel{\sin 3x} \cdot \cancel{\sin 3x} \cdot \cancel{3x}} \cdot \frac{x}{x} \cdot \frac{3}{3}$$

$$\frac{2 \cdot \cancel{\sin x} \cdot \cancel{3x}}{\cancel{\sin 3x} \cdot 3 \cdot \cancel{3x}} \Rightarrow \frac{2}{9}$$

$$\lim_{x \rightarrow 1} \frac{\sin(x^2 - 1)}{x - 1}$$

$$\lim_{x \rightarrow 1}$$

EJERCICIOS CERTAMEN

1.- CALCULAR

$$\lim_{x \rightarrow +\infty} \frac{3x+4}{x+\sqrt{2x^2-5}} =$$

$$\lim_{x \rightarrow -\infty} \frac{3x+4}{x+\sqrt{2x^2-5}} =$$

2.- CALCULAR

$$g(x) = \begin{cases} x+5; & x \leq -3 \\ \sqrt{9-x^2}; & -3 < x < 3 \\ 3-x; & x \geq 3 \end{cases}$$

$$\lim_{x \rightarrow 3} g(x), \lim_{x \rightarrow -3} g(x), \lim_{x \rightarrow 0} g(x)$$

3.- $\lim_{x \rightarrow 1} (1+x^2)^{\cos(\pi x)}$: Repasar el apunte de límites notables

4.- ALGUNA SUCESIÓN

Hacer la mayor cantidad de ejercicios

Suerte

1.- CALCULAR

$$\lim_{x \rightarrow +\infty} \frac{3x+4}{x+\sqrt{2x^2-5}} =$$

$$\lim_{x \rightarrow -\infty} \frac{3x+4}{x+\sqrt{2x^2-5}} =$$

$$1) \lim_{x \rightarrow +\infty} \frac{3x+4}{x+\sqrt{2x^2-5}} \Rightarrow \lim_{x \rightarrow +\infty} \frac{x(3+4/x)}{x(1+\sqrt{2-5/x})} \Rightarrow \frac{3}{1+\sqrt{2}} \Rightarrow 3-3\sqrt{2}$$

$$\lim_{x \rightarrow -\infty} \frac{-3x+4}{-x+\sqrt{2x^2-5}} \Rightarrow \frac{-x(3-4/x)}{-x(1-\sqrt{2-5/x})} \Rightarrow \frac{3}{1-\sqrt{2}} \Rightarrow -3-3\sqrt{2}$$

2.- CALCULAR

$$g(x) = \begin{cases} x+5; & x \leq -3 \\ \sqrt{9-x^2}; & -3 < x < 3 \\ 3-x; & x \geq 3 \end{cases}$$

$$\lim_{x \rightarrow -3} g(x), \lim_{x \rightarrow +3} g(x), \lim_{x \rightarrow 0} g(x)$$

$$\lim_{x \rightarrow -3^-} -3+5 = 2$$

$$\lim_{x \rightarrow -3^+} \sqrt{9-9} = 0$$

$$\lim_{x \rightarrow 3^-} \sqrt{9-9} = 0$$

$$\lim_{x \rightarrow 3^+} 3-3 = 0$$

$$\lim_{x \rightarrow 0} \sqrt{9} = 3$$

$$\lim_{x \rightarrow 3} g(x) = 0$$

$$\lim_{x \rightarrow -3} g(x) = \cancel{2}$$

$$\lim_{x \rightarrow 0} g(x) = 3$$

3.- $\lim_{x \rightarrow 1} (1 + x^2)^{\cos(\pi x)}$: Repasar el apunte de límites notables

$$\lim_{x \rightarrow 1} (1 + x^2)^{\cos(\pi x)}$$

$$\lim_{x \rightarrow 1} 2^{-1} \Rightarrow \frac{1}{2}$$

EJERCICIOS REPASO

$$\begin{aligned} -3 &= 9a - 4 \\ \frac{1}{9} &= a \\ \frac{1}{9} &= a \end{aligned}$$

1. Si $f(x) = \begin{cases} ax^2 - 4 & \text{si } x > 3 \\ -x, & \text{si } x < 3 \end{cases}$ Calcula el valor de **a** para que $\lim_{x \rightarrow 3} f(x)$, exista.

$$\begin{aligned} 8 &= 12a - 4 \\ 12 &= 12a \\ 1 &= a \end{aligned}$$

2. Si $f(x) = \begin{cases} 3ax^2 - 4 & \text{si } x > 2 \\ 4x, & \text{si } x < 2 \end{cases}$ Calcula el valor de **a** para que $\lim_{x \rightarrow 2} f(x)$ exista.

$$\begin{aligned} 5 &= 5a - 3 \\ 8 &= 5a \\ \frac{8}{5} &= a \end{aligned}$$

3. Si $f(x) = \begin{cases} 5ax^2 - 3, & \text{si } x > -1 \\ -5x, & \text{si } x < -1 \end{cases}$ Calcula el valor de **a** para que $\lim_{x \rightarrow -1} f(x)$ exista.

Evaluar los siguientes límites:

1. $\lim_{x \rightarrow 2} \frac{x^2 - 3x + 6}{5x - 1} = \frac{4}{9}$

2. $\lim_{x \rightarrow \pi} \cos 3x$

3. $\lim_{x \rightarrow 4} \sqrt[3]{x+4} = 2$

4. $\lim_{x \rightarrow 4} \sqrt{\frac{x_2 - 1}{x - 1}}$

5. $\lim_{x \rightarrow 0} \frac{4 - \sqrt{16 + x}}{x} = \infty$

6. $\lim_{x \rightarrow 1} \frac{x^3 - 2x^2 + x}{x - 1}$

7. $\lim_{x \rightarrow 4} \sqrt{\frac{25 - (x+1)^2}{5 + (x+1)}} = \sqrt{\frac{0}{25}}$

8. $\lim_{x \rightarrow \frac{\pi}{2}} \sin(x - a)$

9. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin 2x + \cos 2x}{-1} = \frac{0}{-1}$

10. $\lim_{x \rightarrow 3} \frac{(x^2 - 9)(x^3 + 2x^2 - 3x)}{x^2 - 3x}$

1. Calcula los siguientes límites, eliminando las indeterminaciones que se presenten

a) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} = \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \frac{3}{2}$ b) $\lim_{m \rightarrow 1} \frac{3m^2 - 3}{m - 1}$ c) $\lim_{t \rightarrow -4} \frac{t^3 + 64}{t + 4}$

d) $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x^3 - 8}$ e) $\lim_{t \rightarrow 3} \frac{t^2 - 9}{t^2 - 5t + 6}$ f) $\lim_{x \rightarrow 64} \frac{x - 64}{\sqrt{x} - 8}$

g) $\lim_{u \rightarrow 0} \frac{5u^3 + 8u^2}{3u^4 - 16u^2}$ h) $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{x - 1}$ i) $\lim_{x \rightarrow -1} \frac{x^2 + 2x + 1}{x + 1}$

$$j) \lim_{v \rightarrow 3} \frac{\sqrt{v+1} - 2}{v-3}$$

$$k) \lim_{n \rightarrow 0} \frac{\sqrt{5+n} - \sqrt{5}}{\sqrt{2n}}$$

$$l) \lim_{x \rightarrow 2} \frac{x-2}{x^2+x-6}$$

$$m) \lim_{h \rightarrow 3} \frac{\sqrt{2h+3} - h}{h-3}$$

$$n) \lim_{x \rightarrow 2} \frac{(x-2)^2}{x^2-4}$$

$$o) \lim_{x \rightarrow 2} \frac{-x+2}{4-x^2}$$

$$p) \lim_{r \rightarrow 8} \frac{\sqrt[3]{r} - 2}{r-8}$$

$$q) \lim_{x \rightarrow -1} \frac{(x+1)^3}{x^3+1}$$

$$r) \lim_{x \rightarrow 27} \frac{\sqrt[3]{x} - 3}{x-27}$$

LÍMITES DE FUNCIONES TRIGONOMÉTRICAS

Evaluar los siguientes límites, aplicando las fórmulas anteriores, cuando sea necesario, y considerando algunas identidades trigonométricas cuando se requiera:

$$1. \lim_{\theta \rightarrow 0} \frac{\text{Sen} 2\theta}{\theta} \cdot \frac{2}{2} \Rightarrow 2$$

$$2. \lim_{\theta \rightarrow 0} \frac{\text{Sen} \theta}{2\theta} = \frac{1}{2}$$

$$3. \lim_{\theta \rightarrow 0} \frac{\text{Sen} 3\theta}{5\theta} \cdot \frac{3}{3} \Rightarrow \frac{3}{5}$$

$$4. \lim_{\theta \rightarrow 0} \frac{\text{Sen}^2 \theta}{\theta} = \frac{\text{Sen} \theta}{\theta} \cdot \text{Sen} \theta \Rightarrow 0$$

$$5. \lim_{\theta \rightarrow 0} \frac{\text{Sen} 4\theta}{\theta} \cdot \frac{4}{4} = 4$$

$$6. \lim_{\theta \rightarrow 0} \frac{\theta}{\text{Sen} \theta}$$

$$7. \lim_{\theta \rightarrow 0} \frac{\text{Sen} 3\theta}{2\theta} \cdot \frac{3}{3} \Rightarrow \frac{3}{2}$$

$$8. \lim_{x \rightarrow 0} \frac{\text{Sen} 4x}{3x} \cdot \frac{4}{4} \Rightarrow \frac{4}{3}$$

$$9. \lim_{\theta \rightarrow 0} \frac{\text{Sen} \theta / 2}{\theta} \cdot \frac{1/2}{1/2} \Rightarrow \frac{1}{2}$$

$$10. \lim_{\theta \rightarrow 0} \frac{4 \text{Sen} 9\theta}{3\theta} \cdot \frac{3}{3} \Rightarrow 12$$

$$11. \lim_{\theta \rightarrow 0} \frac{\theta}{\text{Sen} 2\theta} \cdot \frac{1/2\theta}{1/2\theta} \Rightarrow \frac{\theta}{2\theta} \Rightarrow \frac{1}{2}$$

$$12. \lim_{\theta \rightarrow 0} \frac{\text{Sen} 7\theta}{4\theta} \cdot \frac{7}{7} \Rightarrow \frac{7}{4}$$

$$13. \lim_{\theta \rightarrow 0} \text{Cos} \theta \frac{\text{Sen} \theta}{\theta} \Rightarrow 1$$

$$14. \lim_{\theta \rightarrow 0} \frac{4 \text{Sen} 25\theta}{3\theta} \cdot \frac{25}{25} \Rightarrow \frac{100}{3}$$

$$15. \lim_{\theta \rightarrow 0} \frac{\pi \text{Sen} \pi \theta}{4 \cdot 2\theta} \Rightarrow \frac{\pi}{8} \frac{\text{Sen} \theta}{2\theta} \cdot \frac{2}{2} \Rightarrow \frac{2\pi}{16} = \frac{\pi}{8}$$

$$16. \lim_{\theta \rightarrow 0} \frac{\text{Cos} \theta}{1 - \text{Sen} \theta} \Rightarrow 1$$

$$17. \lim_{\theta \rightarrow 0} \frac{\text{Tan} \theta}{\theta} \Rightarrow 1$$

$$18. \lim_{\theta \rightarrow 0} \frac{\theta + \text{tan} \theta}{\text{Sen} \theta} \cdot \frac{1/\theta}{1/\theta} \Rightarrow \frac{\theta/\theta + \text{tan} \theta/\theta}{\text{Sen} \theta/\theta} \Rightarrow 2$$

$$19. \lim_{\theta \rightarrow 0} \frac{\sqrt{1 - \cos^2 \theta}}{\theta}$$

$$20. \lim_{\theta \rightarrow 0} \frac{\sqrt{1 - \cos^2 2\theta}}{4\theta}$$

$$21. \lim_{\theta \rightarrow 0} \frac{1 - \cos 2\theta}{2\theta}$$

$$22. \lim_{\theta \rightarrow 0} \frac{1 - \cos 6\theta}{\theta}$$

$$23. \lim_{\theta \rightarrow 0} \frac{3 - 3\cos 2\theta}{6\theta}$$

$$24. \lim_{\theta \rightarrow 0} \frac{5 - 5\cos 3\theta}{15\theta}$$

$$25. \lim_{\theta \rightarrow 0} \frac{5 - 5\cos 4\theta}{10\theta}$$

$$26. \lim_{\alpha \rightarrow 0} \frac{7 - 7\cos 3\alpha}{21\alpha}$$

$$27. \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{2\theta^2}$$

$$28. \lim_{\alpha \rightarrow 0} \frac{4 - 4\sqrt{1 - \sin^2 4\alpha}}{4\alpha}$$

$$29. \lim_{\alpha \rightarrow 0} \frac{8 - 4\sqrt{4 - 4\sin^2 3\alpha}}{9\alpha}$$

$$30. \lim_{\alpha \rightarrow 0} \frac{-4\cos \alpha + 4}{8\alpha}$$

$$31. \lim_{x \rightarrow 0} \frac{\sin x}{x^{1/2}}$$

$$32. \lim_{x \rightarrow 0} \frac{1}{x} \sin \frac{x}{3}$$

$$33. \lim_{x \rightarrow 0} \frac{\sin 2x}{x \cos 3x}$$

$$34. \lim_{x \rightarrow 0} x \sec x \csc x$$

$$35. \lim_{x \rightarrow 0} x \cot 3x$$

$$36. \lim_{x \rightarrow 0} x^2 \csc 2x \cot 2x$$

$$37. \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$38. \lim_{x \rightarrow 0} \sin\left(x + \frac{\pi}{2}\right)$$

$$39. \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\frac{\pi}{2} - x} \quad (\text{sugerencia } u = \frac{\pi}{2} - x)$$

$$40. \lim_{x \rightarrow a} \frac{\cos x - \cos a}{x - a}$$

(sugerencia $u = x - a$)

$$41. \lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi} \quad (\text{sugerencia } u = x - \pi)$$

$$42. \lim_{x \rightarrow 0} \frac{1 - \cos 3x}{\sin x}$$

$$43. \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$44. \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{\pi}{2} - x}{\cos x} \quad (\text{sugerencia } u =$$

$$x - \frac{\pi}{2})$$

RESOLVER

$$37. \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$39. \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\pi - x} \quad (\text{sugerencia } u = \frac{\pi}{2} - x)$$

(sugerencia $u = x - a$)

$$41. \lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi} \quad (\text{sugerencia } u = x - \pi)$$

$$43. \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$x - \frac{\pi}{2})$$

$$37) \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$\frac{\cos(x) \cos(h) - \sin(x) \sin(h) - \cos x}{h}$$

$$\frac{\cos x (\cos h - 1) - \sin x \sin h}{h}$$

$$\frac{\cos x (\cos h - 1)}{h} - \sin x //$$

SON la MISMA
wea

$$39) \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\frac{\pi}{2} - x} \quad \begin{matrix} u = \frac{\pi}{2} - x \\ x = \frac{\pi}{2} - u \\ u \rightarrow 0 \end{matrix} \Rightarrow \lim_{u \rightarrow 0} \frac{1 - \sin(\frac{\pi}{2} - u)}{u} \Rightarrow 1 - \left(\frac{1}{\sin \frac{\pi}{2}} \cos u - \frac{0}{\cos \frac{\pi}{2}} \sin u \right)$$

$$\lim_{u \rightarrow 0} \frac{1 - \cos u}{u} \Rightarrow 0$$

$$41) \lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi} \Rightarrow \begin{matrix} u = x - \pi \\ x = u + \pi \\ u \rightarrow 0 \end{matrix} \lim_{u \rightarrow 0} \frac{\sin(u + \pi)}{u} \Rightarrow \frac{\sin u \cos \pi + \cos u \sin \pi}{u} = \frac{-\sin u}{u} = -1$$

$$16) \lim_{x \rightarrow 0} \frac{\cos x}{1 - \sin x} \cdot \frac{\frac{1}{x}}{\frac{1}{x}}$$

$$\frac{\frac{\cos x}{x}}{\frac{1 - \sin x}{x}}$$

$$\frac{\frac{\cos x}{x}}{\frac{1}{x} - 1} \Rightarrow \frac{\frac{\cos x}{x}}{\frac{1 - x}{x}} \Rightarrow \frac{\cos x}{1 - x} \Rightarrow 1$$

$$1. \lim_{x \rightarrow \infty} \left(\frac{2x+4}{3x+5} \right)^{x-3} \Rightarrow \lim_{x \rightarrow \infty} \left(\frac{\frac{2x}{x} + \frac{4}{x}}{\frac{3x}{x} + \frac{5}{x}} \right)^{x-3} \Rightarrow \left(\frac{2}{3} \right)^{x-3} \Rightarrow \left(\frac{2}{3} \right)^{\infty} \Rightarrow a^{\infty} = 0$$

$$2. \lim_{x \rightarrow 3^+} \left(\frac{1+3x}{2+3x} \right)^{\frac{x^2-9}{x-3}} \Rightarrow \left(\frac{10}{11} \right)^{\frac{(x+3)(x-3)}{x-3}} \Rightarrow \left(\frac{10}{11} \right)^6$$

$$3. \lim_{x \rightarrow 2} (x-1)^{\frac{1}{x-2}} \Rightarrow \frac{1}{e^{x-2}} \cdot (x-1-1) \Rightarrow \frac{x-2}{e^{x-2}} = e$$

Solución.

Es una indeterminación del tipo 1^{∞} LÍMITE IMPORTANTE

$$\lim_{x \rightarrow 2} (x-1)^{\frac{1}{x-2}} = e^{\lim_{x \rightarrow 2} (x-1-1) \cdot \frac{1}{x-2}} = e^1 = e$$

EJEMPLOS IMPORTANTES

Ejemplo 13. $\lim_{x \rightarrow 1} (1+x^2)^{\cos(\pi x)} = 1$. En efecto,

$$\lim_{x \rightarrow 1} (1+x^2)^{\cos(\pi x)} = [\lim_{x \rightarrow 1} (1+x^2)]^{\lim_{x \rightarrow 1} \cos(\pi x)} = 1^{\cos \pi} = 1^{-1} = 1.$$

Ejemplo 14. $\lim_{x \rightarrow 0} (\cos x)^{-1/x^2} = \sqrt{e}$. En efecto, en este caso, $f(x) = \cos x \rightarrow 1$ ($x \rightarrow 0$) y $g(x) = -1/x^2 \rightarrow \infty$ ($x \rightarrow 0$). Tenemos

$$\begin{aligned} \lim_{x \rightarrow 0} [g(x)(f(x) - 1)] &= \lim_{x \rightarrow 0} \left[-\frac{1}{x^2} \cdot (\cos x - 1) \right] \\ &= -\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} \\ &= -\lim_{x \rightarrow 0} \frac{-\frac{x^2}{2}}{x^2}, \text{ ya que } \cos x - 1 \sim -\frac{x^2}{2} \text{ (} x \rightarrow 0 \text{)} \\ &= -\lim_{x \rightarrow 0} \frac{-x^2}{2x^2} = -\left(-\frac{1}{2}\right) \lim_{x \rightarrow 0} \frac{x^2}{x^2} = \frac{1}{2}. \end{aligned}$$

Por lo tanto, de (30) se obtiene

$$\lim_{x \rightarrow 0} (\cos x)^{-1/x^2} = \exp(\lim_{x \rightarrow 0} [g(x)(f(x) - 1)]) = \exp(1/2) = e^{1/2} = \sqrt{e}.$$

SUCESIONES

$$\begin{aligned} \lim_{x \rightarrow 0} (\cos x)^{-\frac{1}{x^2}} &\Rightarrow \lim_{x \rightarrow 0} \frac{1}{x^2} \cdot \frac{1}{\cos x} - 1 \\ &= \frac{1}{x^2} \cdot \frac{1 - \cos x}{\cos x} \\ &= \frac{1 - \cos x}{x^2 \cdot \cos x} \\ &= \frac{1 - \cos x}{x} \cdot \frac{1}{x \cos x} \end{aligned}$$

Problema I. (20 puntos) Considere la sucesión:

$$a_n = \frac{1+2n}{2n-1}, n \in \mathbb{N}$$

a) Determine si a_n es monótona.

b) Muestre que a_n es acotada y convergente. Indique las cotas.

Solución:

a) Monotonía :

$$a_n - a_{n+1} = \frac{1+2n}{2n-1} - \frac{1+2(n+1)}{2(n+1)-1} \quad (3 \text{ puntos})$$

$$a_n - a_{n+1} = \frac{1+2n}{2n-1} - \frac{3+2n}{2n+1} = \frac{(1+2n)(2n+1) - (3+2n)(2n-1)}{(2n-1)(2n+1)} \quad (2 \text{ puntos})$$

$$a_n - a_{n+1} = \frac{4}{(2n-1)(2n+1)} > 0 \quad \forall n \in \mathbb{N} \quad (2 \text{ puntos})$$

En consecuencia, a_n es monótona decreciente estricta. (3 puntos)

b) Convergencia:

$$\lim_{n \rightarrow \infty} \frac{1+2n}{2n-1} = \lim_{n \rightarrow \infty} \frac{n(\frac{1}{n}+2)}{n(2-\frac{1}{n})} = \lim_{n \rightarrow \infty} \frac{(\frac{1}{n}+2)}{(2-\frac{1}{n})} = 1 \quad (3 \text{ puntos})$$

Por lo tanto, a_n converge a 1.

Acotamiento:

Por teorema de convergencia, como a_n es convergente es acotada. (3 puntos)

Dado que a_n es monótona decreciente $a_1 = 3$ es la cota superior y $\lim_{n \rightarrow \infty} \frac{1+2n}{2n-1} = 1$ es la cota inferior. (4 puntos)

Y BUSCAR OTROS DE LIBROS

MUCHA SUERTE

$$a_n = \frac{1+2n}{2n-1}, n \in \mathbb{N}$$

$$a_1 = 3, a_2 = \frac{5}{3}$$

a) Determine si a_n es monótona.

b) Muestre que a_n es acotada y convergente. Indique las cotas.

$$a) \quad a_{m+1} = \frac{2m+2}{2m+1}$$

$$\frac{1+2m}{2m-1} > \frac{2m+2}{2m+1}$$

$$2m + 4m^2 + 1 + 2m > 4m^2 + 4m - 2m - 2$$

$$2m > -3 \quad \checkmark$$

Es monotonía Dec.

$$\lim_{m \rightarrow \infty} \frac{1+2m}{2m-1} \Rightarrow \frac{\frac{1}{m} + \frac{2m}{m}}{\frac{2m}{m} - \frac{1}{m}} \Rightarrow 1 \quad \text{converge}$$

$\therefore$ Por teorema si es monotonía y converge entonces es acotada

$$a_1 = 3$$

$$\text{cota sup: } 3$$

$$\text{cota inf: } 1$$

$$f(x) = \frac{1-2x}{3+x} ; \quad \frac{f(x+h) - f(x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{\frac{1-2(x+h)}{3+(x+h)} - \frac{1-2x}{3+x}}{h} \Rightarrow \frac{\frac{1-2x-2h}{3+(x+h)} - \frac{1-2x}{3+x}}{h} \Rightarrow \frac{(1-2x-2h)(3+x) - (1-2x)(3+(x+h))}{(3+(x+h))(3+x)h} =$$

$$\frac{\cancel{3} - \cancel{6}x - 6h + \cancel{x} - \cancel{2x^2} - 2hx - (\cancel{3} + x + h - \cancel{6}x - \cancel{2x^2} - 2xh)}{(3+x+h)(3+x)h} = \frac{-7h}{\cancel{h}(x^2 + 6x + 9 + 3h + xh + 9)} \Rightarrow \frac{-7}{x^2 + 6x + 9}$$

$$1) \frac{\sqrt{x^2+2}}{x}$$

$$A.V \Rightarrow x=0$$

$$A.H \Rightarrow \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{x^2}{x^2} + \frac{2}{x^2}}}{\frac{x}{x}} \Rightarrow 1, \lim_{x \rightarrow \infty} -1$$

No hay oblicua

$$2) \frac{2x^3 + x^2 - 3x}{2x^2 - 3x - 2}$$

$$A.V \Rightarrow \frac{+3 \pm \sqrt{9 - 4 \cdot 2 \cdot -2}}{4} \Rightarrow \frac{+3 \pm 5}{4} \Rightarrow -\frac{1}{2} \wedge 2$$

$$\text{Asint V. } x = -\frac{1}{2} \text{ y } x = +2$$

A.H $\Rightarrow$ No hay

$$A.O \Rightarrow \lim_{x \rightarrow \infty} \frac{2x^3 + x^2 - 3x}{2x^3 - 3x^2 - 2x} \Rightarrow 1$$

$$\frac{2x^3 + x^2 - 3x}{2x^2 - 3x - 2} - x \Rightarrow \frac{\cancel{2x^3} + x^2 - 3x - \cancel{2x^3} + 3x^2 + 2x}{2x^2 - 3x - 2} \Rightarrow 2$$

$$y = x + 2$$

$$\text{II)} \quad f(x) = \begin{cases} \frac{\cos(x)-1}{ax} & x < 0 \\ \frac{x^2}{a} - b & 0 \leq x \leq 3 \\ \frac{\sqrt{x+13}-4}{4-\sqrt{3x+7}} & x > 3 \end{cases}$$

- a) calcular a y b ; b) Det. Intervalos donde f es continua $f(x) = \frac{1}{\sqrt{x-2}}$ y $g(x) = x+3$
 c) que valores de x $h(x) = \frac{\cos x}{2x-1}$ es discontinua

$$\left. \begin{aligned} \lim_{x \rightarrow 0^-} \frac{\cos(x)-1}{ax} &\Rightarrow \frac{-(1-\cos x)}{ax} \Rightarrow 0 \\ \lim_{x \rightarrow 0^+} \frac{x^2}{a} - b &\Rightarrow \frac{x^2 - ab}{a} = -b \end{aligned} \right\} b=0$$

$$\lim_{x \rightarrow 3} = \frac{9-ab}{a}$$

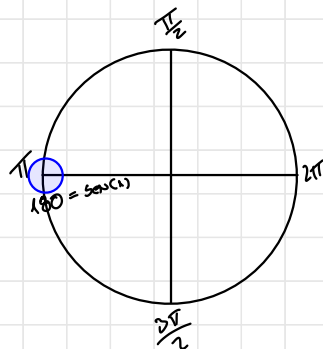
$$\lim_{x \rightarrow 3^+} \frac{\sqrt{x+13}-4}{4-\sqrt{3x+7}} \cdot \frac{\sqrt{x+13}+4}{\sqrt{x+13}+4} \cdot \frac{4+\sqrt{3x+7}}{4+\sqrt{3x+7}} \Rightarrow \frac{(x+13-16)(4+\sqrt{3x+7})}{(16-3x-7)(\sqrt{x+13}+4)}$$

$$\Rightarrow \frac{(x-3)(4+\sqrt{3x+7})}{(-3x+9)(\sqrt{x+13}+4)} \Rightarrow \frac{(x-3)(4+\sqrt{3x+7})}{-3(x-3)(\sqrt{x+13}+4)} \Rightarrow \frac{8}{-24} \Rightarrow -\frac{1}{3}$$

$$\frac{9-ab}{a} = -\frac{1}{3} \Rightarrow 24 = -a \Rightarrow a = -24$$

b) $\frac{1}{\sqrt{x+3-2}} \Rightarrow \frac{1}{\sqrt{x+1}} \Rightarrow x+1 > 0 \Rightarrow x > -1 \Rightarrow \text{Para que sea cont. }]-1, +\infty[$

c) $\frac{\cot x}{2x-1} \Rightarrow \frac{\frac{\cos x}{\sin x}}{2x-1} \Rightarrow \frac{\cos x}{2x \sin x \cdot \sin x} \Rightarrow \frac{\cos x}{\sin x(2x-1)} \Rightarrow x = \frac{1}{2} \text{ y } x = \frac{1}{2}$



$$3) \frac{3e^x/\sqrt{x} + \sqrt{x} \sin x}{\cos x} + \left(\frac{3e^x/\sqrt{x} + \frac{3e^x}{x}}{x} + \left(\frac{1}{2\sqrt{x}} \sin x + \sqrt{x} \cos x \right) \right)$$

$$\frac{3e^x/\sqrt{x} + 3e^x}{x} + \frac{\sin x + 2x \cos x}{2\sqrt{x}}$$

$$\frac{3e^x \left(\frac{1}{\sqrt{x}} + 1 \right)}{x} + \frac{\sin x + 2x \cos x}{2\sqrt{x}}$$

$$\frac{1-2^x}{\cos x}$$

$$\frac{(1-2^x)'(\cos x) - (1-2^x)(\cos x)'}{\cos^2 x} \quad (y = a^x \Rightarrow y' = a^x \cdot \ln(a))$$

Det equation. Rect. tg $y = \frac{2x}{(x+1)^2}$, $x=0$

$$y' = \frac{2 \cdot (x+1)^2 - (2x \cdot 2(x+1) \cdot 1)}{(x+1)^4} = \frac{2x^2 + 4x + 2 - (4x^2 + 4x)}{(x+1)^4} = \frac{-2x^2 + 2}{(x+1)^4} \Rightarrow x=0 \Rightarrow 2$$

$m=2$; si $x=0 \rightarrow y=0$

$$y = 2x$$

$$y = \sqrt{\sin(x^2+1)} \quad , \quad y''?$$

$$y' = \frac{1}{2\sqrt{\sin(x^2+1)}} \cdot \cos(x^2+1) \cdot 2x \Rightarrow \frac{x \cos(x^2+1)}{\sqrt{\sin(x^2+1)}}$$

$$y'' = \frac{(\cos(x^2+1) + x \cdot -\sin(x^2+1) \cdot 2x) \left(\sqrt{\sin(x^2+1)} \right) - (x \cos(x^2+1)) \left(\frac{1}{2\sqrt{\sin(x^2+1)}} \cdot \cos(x^2+1) \cdot 2x \right)}{\sin(x^2+1)}$$

$$\frac{(\cos(x^2+1) - 2x^2 \sin(x^2+1)) \left(\sqrt{\sin(x^2+1)} \right) - (x \cos(x^2+1)) \left(\frac{x \cos(x^2+1)}{\sqrt{\sin(x^2+1)}} \right)}{\sin(x^2+1)}$$

$$\cos(xy) = e^{x^2}$$

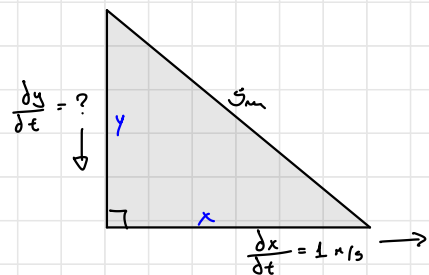
$$, y = e^{\pi-1}$$

$$y' = 0$$

$$, P(1, \pi)$$

$$-\sin(xy) \cdot y + x y' = e^{x^2} \cdot 2x$$

$$y' = \frac{e^{x^2} \cdot 2x}{\sin(xy)y}$$



$$x^2 + y^2 = 25, \text{ cuando } x = 3 \Rightarrow y^2 = 16$$

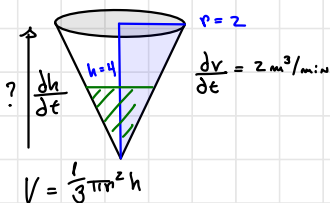
$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

$$y = 4$$

$$2 \cdot 3 \cdot 1 + 2 \cdot 4 \cdot \frac{dy}{dt} = 0$$

$$6 + 8 \frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -\frac{6}{8} \Rightarrow -\frac{3}{4} \text{ m/s}$$



Dejar todo en h

$$\frac{r}{h} = \frac{2}{4} \Rightarrow r = \frac{h}{2}$$

$$V = \frac{\pi}{3} \cdot \frac{h^3}{4}$$

$$V = \frac{\pi}{12} \cdot h^3$$

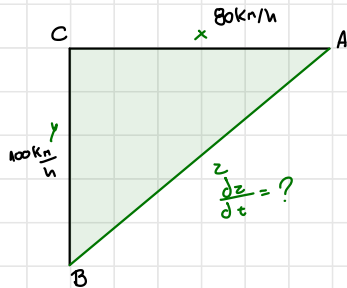
$$\frac{dV}{dt} = \frac{\pi}{12} \cdot 3h^2 \cdot \frac{dh}{dt}$$

$$2 \text{ m}^3/\text{min} = \frac{\pi}{12} \cdot 3 \cdot 3 \text{ m}^2 \cdot \frac{dh}{dt}$$

$$2 \text{ m}^3/\text{min} = \frac{\pi}{4} \cdot \frac{dh}{dt}$$

$$\frac{8 \text{ m}^3/\text{min}}{\pi} = \frac{dh}{dt}$$

$$\frac{8 \text{ m}^3/\text{min}}{\pi} = \frac{dh}{dt}$$



cuando $A = 0,3 \text{ km}$ $\frac{3}{10}$
 cuando $B = 0,4 \text{ km}$ $\frac{4}{10}$

$$z^2 = x^2 + y^2 \rightarrow z = 0,5 \rightarrow \frac{5}{10}$$

$$2z \cdot \frac{dz}{dt} = zx \cdot \frac{dx}{dt} + zy \cdot \frac{dy}{dt}$$

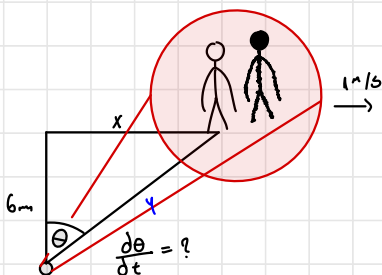
$$2 \cdot \frac{5}{10} \cdot \frac{dz}{dt} = 2 \cdot \frac{3}{10} \cdot 80 \text{ km/h} + 2 \cdot \frac{4}{10} \cdot 100 \text{ km/h}$$

$$\frac{dz}{dt} = \frac{3}{5} \cdot 80 + \frac{4}{5} \cdot 100$$

$$\frac{dz}{dt} = -48 + (-80)$$

$$\frac{dz}{dt} = -128 \text{ km/h}$$

Se acercan a una vel. de 128 km/h



$$\frac{x}{6} = \tan \theta \rightarrow x = 6 \tan \theta ; \text{ cuando } x = 4,5 \Rightarrow y = 7,5 \Rightarrow \cos \theta = \frac{4}{5}$$

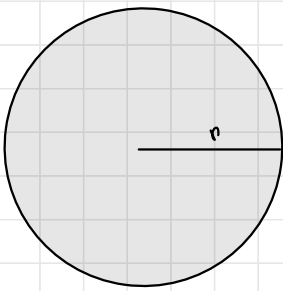
$$\frac{dx}{dt} = 6 \sec^2 \theta \cdot \frac{d\theta}{dt}$$

$$\frac{1}{6} \cos^2 \theta = \frac{d\theta}{dt}$$

$$\frac{1}{6} \cos^2 \theta = \frac{d\theta}{dt}$$

$$\frac{1}{6} \cdot \left(\frac{4}{5}\right)^2 = \frac{d\theta}{dt}$$

$$\frac{16}{6 \cdot 25} = \frac{d\theta}{dt} \text{ rad/s}$$



$$\frac{dr}{dt} = 2 \text{ m/s}$$

$$\frac{dA}{dt} = ?$$

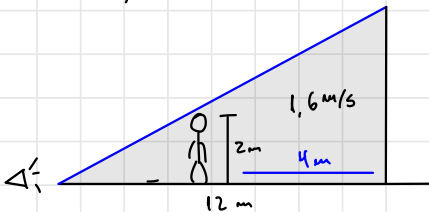
$$r = 30 \text{ m}$$

$$A = \pi r^2$$

$$\frac{dA}{dt} = \pi \cdot 2r \cdot \frac{dr}{dt}$$

$$\frac{dA}{dt} = \pi \cdot 2 \cdot 30 \cdot 2 \text{ m/s}$$

$$\frac{dA}{dt} = 120\pi \text{ m}^2/\text{s}$$



Dist. hombre edificio $\Rightarrow x(t) = 4 \text{ m}$

Altura sombra hombre $= h(t)$

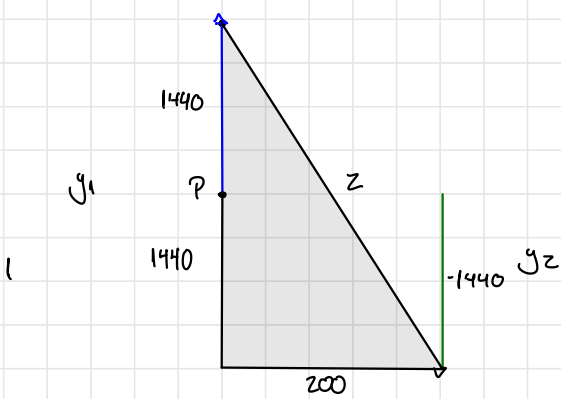
incógnita $= \frac{dh}{dt}$

Triángulos: $\frac{h(t)}{2} = \frac{12}{x(t)}$ $\rightarrow \frac{\text{Altura triángulo grande}}{\text{Altura triángulo pequeño}} = \frac{\text{base triángulo grande}}{\text{base triángulo pequeño}}$

$$h(t) = \frac{24}{x(t)}$$

$$\frac{dh}{dt} = \frac{-24 \cdot \frac{dx}{dt}}{x(t)^2}$$

$$\frac{dh}{dt} = \frac{-24 \cdot 1.6 \text{ m/s}}{16} = 2.4 \text{ m/s} \quad * \text{Debido a que decrece}$$



$1 \text{ min} \rightarrow 60 \text{ seg}$
 $5 \text{ min} \rightarrow 900 \text{ seg}$
 $20 \text{ min} \rightarrow 1200 \text{ seg}$

$900 \cdot 1,6$
 1440 m

$1200 \cdot 1,2$
 1440 m

$$\sqrt{200^2 + 2880^2} = c$$

$$x \cdot \frac{dx}{d\epsilon} + (y_1 - y_2) \left(\frac{dy_1}{d\epsilon} - \frac{dy_2}{d\epsilon} \right) = z \cdot \frac{dz}{d\epsilon}$$

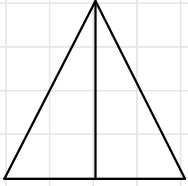
$$\frac{\overset{\text{const}}{200 \cdot 0} + (1440 - (-1440)) \cdot (-(-1,6))}{\sqrt{200^2 + 2880^2}} = \frac{dz}{d\epsilon}$$

$$= 2,79 \text{ m/s}$$

Forma 2:

$$z^2 = x^2 + (y_1(t) - y_2(t))^2$$

$$z^2 = 200^2 + (1,2(t+5) + 1,6t)^2$$



$$\frac{dh}{dt} = 1 \text{ cm/min}$$

$$\frac{dA}{dt} = 2 \text{ cm}^2/\text{min}$$

$$A = \frac{b \cdot h}{2} \Rightarrow 100 \text{ cm}^2 = \frac{b \cdot 10 \text{ cm}}{2} \Rightarrow b = 20 \text{ cm}$$

$$2 \cdot \frac{dA}{dt} = \frac{db}{dt} \cdot h + b \cdot \frac{dh}{dt}$$

$$4 \text{ cm}^2/\text{min} = \frac{db}{dt} \cdot 10 \text{ cm} + 20 \cdot 1 \text{ cm/min}$$

$$\frac{4 \text{ cm}^2/\text{min} - 20 \text{ cm/min}}{10 \text{ cm}} \Rightarrow 1,6 \frac{\text{cm}}{\text{min}}$$